MEMOREASE

AI-Powered Study Companion and Flashcard Learning Platform

USING DESIGN THINKING APPROACH

A MINI PROJECT REPORT FOR THE COURSE

GE 23627 DESIGN THINKING AND INNOVATION

Submitted by

RAGHUL M (240701409)

II YEAR B.E

in

COMPUTER SCIENCE AND ENGINEERING



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

RAJALAKSHMI ENGINEERING COLLEGE

THANDALAM, CHENNAI 602 105

MAY 2026

BONAFIDE CERTIFICATE

Certified that the mini project titled "MEMOREASE" is the Bonafide work carried out by RAGHUL M (240701409) under my supervision. This project has been completed in partial fulfilment of the requirements of the course GE23627 - Design Thinking and Innovation.

Certified further that, to the best of my knowledge and belief, the work reported herein is original and does not form part of any other project, report, or work submitted previously for any academic award.

This project contributes toward achieving the following Sustainable Development Goals (SDGs):

- SDG 4 - Quality Education: Supporting students with free and accessible AI-based learning tools.
- SDG 9 - Industry, Innovation and Infrastructure: Demonstrating an innovative digital education platform.
- SDG 10 - Reduced Inequalities: Reducing the cost barrier created by paid study applications.
- SDG 17 - Partnerships for the Goals: Encouraging open-source learning and community support.

Student Signature with Name

1. RAGHUL M

Signature of the Supervisor with date

Signature Examiner-1 **Signature Examiner-2**

ABSTRACT

MemorEase is an AI-powered study companion designed to help students convert study material into flashcards, reviewers, quizzes, and structured learning resources. The platform is presented as a free and open-source alternative to paid study tools such as Quizlet and Gizmo, with the goal of making effective study support available to students without financial or installation barriers.

The main problem addressed by MemorEase is that many students have large amounts of notes, PDF content, textbook points, and exam material, but they often struggle to organize this information into revision-friendly formats. Traditional note reading is time consuming, while manual flashcard creation can be repetitive. MemorEase solves this by using AI-assisted generation to transform raw study material into learning assets such as cards, reviewer notes, practice tests, and study resources.

The platform includes features such as a materials hub, AI-powered content summarization, flashcards, reviewer generation, quiz mode, Pomodoro timer, achievements, study calendar, leveling system, account settings, Google OAuth login, and privacy-focused storage. The visible changelog and website pages also indicate features such as Supabase authentication, row-level security, Zod validation, hCaptcha protection, rate limiting, PDF/DOCX export, and Google Gemini based AI support.

The project follows the Design Thinking approach. The Empathize stage focuses on student pain points such as exam pressure, expensive tools, unorganized notes, low motivation, and lack of quick revision. The Define stage frames the need for a simple, free, browser-based study platform. The Ideate stage explores AI flashcards, quizzes, summaries, Pomodoro, gamification, calendar, and export options. The Prototype stage is represented by the deployed MemorEase website, and the Test stage evaluates clarity, usability, security, and student usefulness.

MemorEase is valuable because it combines artificial intelligence, educational technology, spaced repetition, productivity support, and open access. The project has strong practical relevance for students and can grow into a larger platform with mobile apps, collaborative study groups, adaptive scheduling, analytics, and deeper AI personalization.

LIST OF TABLES

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	3
	LIST OF TABLES	4
	LIST OF FIGURES	5
	LIST OF ABBREVIATIONS	6
1	INTRODUCTION	7
1.1	DESIGN THINKING APPROACH	9
2	LITERATURE REVIEW	11
3	DOMAIN AREA	14
4	EMPATHIZE STAGE	17
5	DEFINE STAGE	19
6	IDEATION STAGE	21
7	PROTOTYPE STAGE	23
8	TESTING AND FEEDBACK	46
9	RE-DESIGN AND IMPLEMENTATION	48
10	RESULT AND CONCLUSION	49
11	WRITE UPS	50
12	FUTURE WORK	54
13	LEARNING OUTCOMES OF DESIGN THINKING	56
14	REFERENCES	58

LIST OF FIGURES

Figure no	Title
4.1	Empathy Map for MemorEase
5.1	Problem Definition Summary
6.1	Ideation Mind Map
7.1 - 7.21	Prototype User Interface Screens
8.1	Testing and Feedback Summary

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DB	Database
DT	Design Thinking
HTML	HyperText Markup Language
JWT	JSON Web Token
LLM	Large Language Model
MVP	Minimum Viable Product
OAuth	Open Authorization
PDF	Portable Document Format
RLS	Row Level Security
SRS	Spaced Repetition System
UI	User Interface
UX	User Experience
URL	Uniform Resource Locator
XP	Experience Points

1. INTRODUCTION

Education has become more digital than ever before. Students now receive study material in many forms such as PDF files, online notes, presentation slides, typed documents, handwritten summaries, and textbook extracts. Even though the amount of learning material has increased, students still face difficulty in converting that material into a simple revision plan.

Many students prepare for examinations by reading the same notes repeatedly. This method consumes time and does not always improve long-term memory. Research in learning science shows that active recall, spaced repetition, practice testing, and structured review are more effective than passive reading. However, creating flashcards, quizzes, and revision notes manually can be tiring.

MemorEase is developed as a solution to this problem. It is an AI-powered study platform that transforms study material into flashcards, reviewers, quizzes, and other learning tools. The website describes MemorEase as an open-source alternative to Quizlet and Gizmo, with the objective of helping students study smarter and not harder.

The project focuses on accessibility. Students should be able to open the browser, sign in quickly, upload or enter study content, and start generating useful learning assets. The platform is designed to remove barriers such as paid subscriptions, complicated installation steps, and slow manual preparation.

MemorEase is more than a flashcard generator. It includes a complete learning environment with a materials hub, AI summarization, flashcards, review modes, quiz mode, Pomodoro timer, achievements, study calendar, leveling system, account settings, privacy policy, and visible changelog. These features make the project suitable as a complete educational technology prototype.

The project is relevant for students preparing for tests, semester examinations, competitive exams, or daily revision. By combining AI content generation and study workflow tools, MemorEase can reduce preparation time and improve student engagement.

Need for the Project

Students often collect study material from different sources but do not have enough time to convert those materials into revision-friendly formats. This creates stress before examinations because notes remain lengthy and unorganized.

Paid study platforms may provide useful tools, but not every student can afford them. Some tools also require installation, credit cards, or complicated onboarding. MemorEase addresses this need by providing a free browser-based study companion.

Another need is motivation. Students may lose interest when revision feels repetitive. MemorEase includes gamified elements such as achievements, levels, XP, calendar, and interactive study modes so that learning feels more structured and engaging.

The project also supports modern learning habits. Students want quick access, mobile-friendly design, automatic generation, export options, and secure account sync. MemorEase brings these ideas into one platform.

Objectives

- To build a free and accessible AI-powered study platform for students.
- To convert study material into flashcards, reviewers, and practice tests.
- To support active recall and spaced repetition for better memory retention.
- To provide a materials hub where learning resources can be organized.
- To include productivity tools such as Pomodoro timer, calendar, achievements, and leveling.
- To provide Google OAuth login and privacy-focused user account management.
- To design a clean, simple, and responsive user interface.
- To make the project scalable for future mobile and collaborative learning features.

Existing System

In the existing study workflow, students usually read notes, highlight points, and manually prepare question answers or flashcards. This process is time consuming and may not be efficient when the syllabus is large.

Some online study tools support flashcards and quizzes, but many advanced features are locked behind payment plans. This creates an access gap between students who can pay and students who cannot.

Existing tools may also separate productivity features from learning features. A student may need one application for flashcards, another for Pomodoro, another for calendar planning, and another for document summarization. This fragmented workflow makes revision less organized.

Proposed System

MemorEase proposes a single platform where students can transform study materials into structured learning assets. The user can begin from the landing page, sign in using Google, access materials, create flashcards or reviewers, use quiz mode, and manage study sessions.

The proposed system uses AI to reduce manual effort. Instead of typing every flashcard by hand, the student can provide study content and allow the platform to extract concepts, definitions, summaries, and practice questions.

The system is also designed for continuous study. Tools such as flashcards, spaced repetition, Pomodoro timer, achievements, calendar, and leveling encourage repeated engagement. This makes MemorEase useful for long-term learning instead of only last-minute preparation.

Advantages of MemorEase

- Students can generate revision material faster.
- The free and open-source positioning reduces financial barriers.
- AI summarization helps convert lengthy content into key learning points.
- Flashcards and quizzes support active recall.
- Pomodoro timer supports focused study sessions.
- Achievements, levels, and calendar features improve motivation.
- Google OAuth and Supabase based flow simplify account access.
- The browser-based design avoids installation complexity.

1.1 DESIGN THINKING APPROACH

Design Thinking is a human-centered approach used to understand user needs and create practical solutions through iteration. For MemorEase, the user is primarily a student who wants to study efficiently, reduce preparation stress, and access tools without paying for expensive platforms.

The Design Thinking approach is useful because the project is not only about AI generation. It is also about how students feel while studying, how they organize material, how they stay motivated, and how they trust a study platform with their uploaded documents.

MemorEase applies empathy by focusing on the student's real situation: limited time, large syllabus, scattered notes, exam pressure, and need for simple tools. The solution is then defined, ideated, prototyped, and tested through the deployed website.

Stage	Application in MemorEase
Empathize	Understand students, study habits, exam pressure, cost barriers, and revision difficulties.
Define	Frame the need for a free AI study companion that turns material into learning assets.
Ideate	Generate ideas such as AI flashcards, reviewer notes, quiz mode, Pomodoro, achievements, and calendar.
Prototype	Develop the deployed website with landing page, study previews, help pages, and OAuth flow.
Test	Evaluate clarity, usefulness, navigation, security, and student acceptance.

1.2 STANFORD DESIGN THINKING MODULE AND ITS PHASES

The Stanford Design Thinking model includes five major stages: Empathize, Define, Ideate, Prototype, and Test. This model was selected because it is simple, iterative, and suitable for converting a student problem into a working product concept.

In MemorEase, the Empathize stage studies how students currently revise and why existing tools may not be enough. The Define stage identifies the core problem of unorganized study materials and limited access to intelligent tools. The Ideate stage produces the feature set. The Prototype stage is the deployed web application. The Test stage checks whether the workflow is understandable and helpful.

The process is iterative. Feedback from students can be used to redesign the dashboard, improve AI output quality, simplify onboarding, strengthen privacy controls, and make the study modes more useful.

2. LITERATURE REVIEW

The literature review for MemorEase focuses on educational technology, artificial intelligence in learning, flashcard systems, spaced repetition, retrieval practice, gamification, productivity methods, and privacy in cloud-based study tools.

Research on active recall shows that learners remember better when they try to retrieve information rather than only rereading notes. Flashcards are a practical active recall tool because they ask the learner to recall an answer before checking it.

Spaced repetition is another important learning method. Instead of reviewing all material at the same frequency, items are repeated at increasing intervals. Difficult cards are shown more often and easier cards are shown less often. This supports long-term retention and reduces unnecessary repetition.

Practice testing is also helpful because it prepares students for exam-like conditions. MemorEase includes quiz mode and practice test generation, which aligns with this learning principle.

AI in education can help students summarize content, generate questions, create definitions, and organize study material. However, AI-generated content must be reviewed by the learner because errors or oversimplifications can happen. Therefore, editable cards and reviewer notes are important.

Gamification research shows that progress indicators, achievements, points, levels, and streaks can improve motivation when used carefully. MemorEase includes achievements, XP, leveling, and activity calendar ideas, making the study process more engaging.

Pomodoro technique literature supports focused work intervals followed by short breaks. The Pomodoro timer in MemorEase helps students manage attention and avoid burnout during study sessions.

Privacy is important in educational platforms because users may upload personal notes, school documents, and account information. The MemorEase website includes a privacy policy and mentions third-party authentication and AI processing. The changelog also indicates security features such as row-level security, hCaptcha, validation, rate limiting, CSP, and HSTS.

Detailed Review of Related Concepts

Flashcard learning systems are widely used because they convert broad content into small recall units. MemorEase improves this idea by allowing AI to generate cards from study materials, reducing the manual effort required from the student.

Reviewer notes are useful when students need a structured summary instead of only question-answer cards. The website preview shows biology reviewer content such as terms and definitions. This demonstrates how dense lecture notes can be converted into organized summaries.

AI-powered extraction is especially relevant when students upload PDF, DOCX, or plain text content. The About page states that MemorEase works with PDF, DOCX, and plain text. This helps students use existing study material instead of retyping content.

Open-source educational software is important because it allows transparency, community contribution, and free access. MemorEase describes itself as open source and free for students, which makes the project socially useful.

Authentication and cloud storage are important when the platform supports user accounts and saved materials. The login flow redirects to Google sign-in through Supabase, indicating that authentication and user-specific data access are central parts of the system.

Study Area	Relevance to MemorEase
Active Recall	Supports flashcards and quiz-based learning.
Spaced Repetition	Supports repeated review of difficult concepts over time.
AI in Education	Supports automatic generation of cards, reviewers, and tests.
Gamification	Supports achievements, XP, levels, and study engagement.
Pomodoro Technique	Supports focused study sessions with timed breaks.
Privacy and Security	Supports safe handling of uploaded materials and accounts.

Inference from Literature Review

The review shows that MemorEase combines several strong learning principles into one system. Active recall, spaced repetition, practice testing, and focused study sessions are all useful individually, but the project becomes stronger when these ideas are connected inside one platform.

The review also shows that AI should not replace learning. Instead, it should reduce preparation effort and help students create better study material. The student still studies, edits, reviews, and tests knowledge actively.

Another inference is that accessibility matters. A free browser-based tool can help more students compared to paid or installation-heavy tools. This supports the project's positioning as a free alternative to popular study platforms.

Security is a major requirement because uploaded study materials and account data must be protected. The visible changelog mentions security improvements, which shows that the project has considered this issue.

Overall, the literature review supports MemorEase as a meaningful educational technology project because it connects learning science, AI automation, productivity, motivation, and secure digital infrastructure.

3. DOMAIN AREA

The domain area of MemorEase is educational technology. More specifically, it belongs to the field of AI-assisted study tools, flashcard learning systems, digital note summarization, and productivity support for students.

Educational technology is not limited to online classes. It also includes tools that help learners organize content, revise material, test themselves, track progress, and manage study time. MemorEase fits into this domain because it supports the complete revision workflow.

The main users are students. They may be school students, college students, competitive exam aspirants, or self-learners. The platform is designed to work from the browser, which makes it usable across devices.

The domain also includes backend and cloud services. User accounts, uploaded materials, generated content, saved cards, quiz results, calendar records, and achievements require persistent storage and secure access control.

System Modules

Module	Purpose
Landing and Onboarding	Introduces the project, explains value, and guides students to start learning.
Authentication	Uses Google OAuth and Supabase flow to manage user access.
Materials Hub	Stores and organizes study assets such as cards and reviewers.
AI Generation	Transforms study material into flashcards, reviewers, and practice tests.
Flashcard Study	Supports active recall and spaced repetition based revision.
Quiz Mode	Tests knowledge through AI-generated practice questions.
Pomodoro Timer	Helps students study in focused time blocks.
Gamification	Tracks XP, achievements, levels, and activity calendar.
Security Layer	Protects accounts and data using validation, RLS, captcha, and security headers.

Technology Stack

The deployed website is hosted on Vercel and appears to be a modern Next.js based web application. The website pages and routing style indicate a browser-first design with serverless deployment. The visible changelog mentions Supabase, Google OAuth, Gemini client service, Zod validation, hCaptcha, row-level security, rate limiting, CSP, and HSTS.

Because the public repository was not accessible during report preparation, the report uses the deployed website, visible routes, changelog, and provided project analysis as evidence. Any internal folder structure is therefore described as a recommended or inferred structure rather than a confirmed source-code view.

Layer	Technology / Feature
Frontend	Responsive web interface, likely Next.js with modern CSS and component based layout.
Hosting	Vercel deployment, public URL, and static/serverless web delivery.
Authentication	Google OAuth using Supabase authentication callback.
Database / Storage	Supabase indicated by auth URL and changelog references.
AI Service	Gemini client service mentioned in changelog.
Validation	Zod validation mentioned in changelog.
Security	RLS, hCaptcha, CSP, HSTS, rate limiting, and input validation.
Analytics	PostHog mentioned in terms for product analytics.

Data Flow

The expected data flow begins when a student signs in and creates or uploads study material. The platform stores the material under the user's account. The AI service then processes the content and generates flashcards, reviewer notes, or quiz questions.

Generated content is shown in the materials hub. The student can review cards, take quizzes, export resources, and track progress. If spaced repetition is applied, every review updates the next review time and difficulty state of the card.

For productivity features, the Pomodoro timer records focus sessions and can connect to progress statistics. Gamification features such as XP and achievements may update when the student completes study actions.

Database Design

Table / Collection	Main Fields
users	user_id, name, email, provider, created_at, last_login
materials	material_id, user_id, title, type, source_text, file_url, created_at
flashcards	card_id, material_id, term, definition, difficulty, interval, due_date
review_logs	log_id, card_id, user_id, result, reviewed_at, next_due_date
quizzes	quiz_id, material_id, question, options, answer, explanation
pomodoro_sessions	session_id, user_id, focus_minutes, break_minutes, completed_at
achievements	achievement_id, user_id, badge_name, xp_value, unlocked_at

Database Design Explanation

The database design is centered on the relationship between users, materials, generated study content, review activity, and progress tracking. Every major object should be linked to a user account so that students can access their own materials securely after login.

The materials table acts as the main parent record. A material may produce many flashcards, quiz questions, and reviewer sections. This makes the system organized because all generated outputs can be traced back to the uploaded or entered study source.

The flashcards and review logs support spaced repetition. A card stores its current difficulty, interval, and due date, while review logs record actual user performance. This separation helps the platform analyze learning progress over time.

The pomodoro sessions and achievements tables support motivation. They are not required for basic flashcard generation, but they make MemorEase a complete study companion instead of only an AI content generator.

4. EMPATHIZE STAGE

The Empathize stage is used to understand students and their study problems. The target user is a student who has many study materials but limited time to organize them into useful revision resources.

Students may feel overwhelmed when lecture notes are lengthy, PDF content is dense, and exam dates are close. They may also feel frustrated when useful study platforms require payment or complicated setup.

The empathy study for MemorEase focuses on affordability, speed, simplicity, motivation, and trust. Students want to start quickly, generate useful material, revise actively, and feel confident that their uploaded notes are handled safely.

The deployed website directly reflects this empathy: the headline says 'Study smarter not harder', and the supporting text promises transformation of study material into cards, reviewers, and exams instantly.

Empathy Map

SAYS	THINKS
I have notes, but I do not know how to revise them quickly.	Can this tool create useful cards and quizzes without wasting my time?
DOES	FEELS
Uploads or enters study content, checks generated cards, reviews material, and takes quizzes.	Pressured by exams but motivated when progress is visible.
PAIN	GAIN
Expensive tools, scattered notes, low motivation, and manual flashcard creation.	Free access, AI summaries, active recall, Pomodoro focus, and gamified progress.

Fig 4.1

The empathy map shows that the student needs more than a generator. The student needs a complete study experience that reduces effort and increases confidence.

The most important insight is that students are not against studying. They mainly need help converting messy material into organized revision assets. MemorEase addresses this by connecting AI generation with study modes.

User Needs Identified

User	Needs Identified
Student	Generate flashcards, summaries, quizzes, and study plans quickly from existing material.
Self Learner	Use free tools without installation and access content from any device.
Exam Candidate	Practice active recall, track progress, and revise high-priority concepts.
Platform Admin / Maintainer	Maintain security, reliability, AI quality, and user support.

User Personas

Persona	Description	Requirement
College Student	Has PDF notes and wants fast revision before exams.	Needs AI-generated flashcards and reviewers.
Competitive Exam Learner	Studies repeatedly over weeks or months.	Needs spaced repetition, quizzes, calendar, and progress tracking.
Busy Student	Has limited time and needs focused study sessions.	Needs Pomodoro timer and quick summaries.

Outcome of Empathize Stage

The Empathize stage showed that MemorEase should be simple, fast, free, and trustworthy. Students should not need to understand technical AI details before using the platform.

The study process should feel guided. The student should see clear actions such as start learning, create cards, open materials, take quiz, use Pomodoro, and check progress.

The empathy insights helped move the project into the Define stage with a clear problem statement focused on student accessibility and study efficiency.

5. DEFINE STAGE

The Define stage converts the empathy findings into a focused problem statement. MemorEase focuses on the problem that students have study materials but lack a free, fast, and organized way to convert them into active revision tools.

The core problem is not only that students have many notes. The deeper issue is that notes are often passive. Students need active recall resources such as flashcards, quizzes, and practice questions.

Another problem is tool fragmentation. Students may need separate tools for summaries, flashcards, quizzes, Pomodoro, calendars, and progress tracking. MemorEase tries to bring these into one study platform.

The final selected problem statement is: Students need a free, accessible, and AI-powered platform that can transform study materials into organized flashcards, reviewers, quizzes, and productivity tools, helping them revise effectively without financial or technical barriers.

- Students spend too much time converting notes into revision material.
- Paid study platforms create barriers for students who cannot afford subscriptions.
- Manual flashcard creation is slow and repetitive.
- Students need motivation through progress tracking and gamification.
- Uploaded study materials must be handled securely and privately.

Problem Definition Summary

Aspect	Defined Problem
Learning	Students need active recall, spaced repetition, and practice testing.
Accessibility	Students need free browser-based access without installation.
Organization	Study materials need to be stored, categorized, and editable.
Motivation	Students need progress indicators, achievements, and calendar tracking.
Security	Accounts and uploaded materials must be protected.

Fig 5.1

Constraints

The platform must generate useful content quickly, but AI output quality can vary based on the input material. Therefore, generated cards and reviewers should remain editable.

The system must be free for students, but it still needs hosting, AI usage, database storage, and domain support. This creates a sustainability challenge for the platform maintainer.

Privacy is a constraint because study materials may include personal notes or institutional content. Authentication, access control, validation, and security policies are required.

The deployed prototype also has external dependency constraints because it relies on cloud services such as Google OAuth, Supabase, Vercel, and AI processing.

6. IDEATION STAGE

The Ideation stage explored many possible features that could help students study better. The team considered AI summarization, flashcard generation, quiz mode, reviewer notes, Pomodoro timer, calendar, achievements, progress levels, PDF export, DOCX export, sharing, and mobile responsiveness.

Ideas were evaluated based on usefulness, feasibility, student need, and connection to the problem statement. Features that directly reduce study preparation effort were given high priority.

The selected feature set shows a balance between AI automation and learning science. AI helps prepare the material, while flashcards, quizzes, and repetition help the student actually learn it.

Mind Mapping

Main Idea	Sub Ideas
AI Study Generation	Flashcards, reviewers, practice tests, summaries, extraction modes.
Learning Modes	Cards, learn mode, match mode, practice mode, spaced repetition.
Productivity	Pomodoro timer, tasks, focus sessions, reminders.
Motivation	XP, achievements, levels, activity calendar, streaks.
Security	Google OAuth, Supabase RLS, validation, captcha, rate limits.
Accessibility	Free access, browser-based use, responsive design, no installation.

Fig 6.1

Alternative Ideas Considered

One idea was to build only a basic flashcard maker. This would be simple, but it would not solve the wider problem of organizing and motivating study.

Another idea was to build only a quiz generator. This would help with testing, but students would still need separate tools for summaries, cards, and time management.

A third idea was to build a mobile-only application. This could be convenient, but it would add installation barriers. The browser-based approach was selected because it is easier to access instantly.

The final idea combines AI generation, materials management, study modes, Pomodoro, gamification, authentication, and responsive web design into one

platform.

Functional Requirements

- The user should be able to sign in using Google.
- The user should be able to create or upload study material.
- The system should generate flashcards, reviewer notes, and practice questions.
- The user should be able to organize generated material in a materials hub.
- The user should be able to study through flashcards and quizzes.
- The user should be able to use Pomodoro focus sessions.
- The system should track achievements, XP, and activity.
- The system should protect user data through authentication and access control.

Non-Functional Requirements

- The interface should be simple and responsive.
- The platform should load quickly through browser access.
- AI generation should provide useful and readable content.
- The system should validate inputs before processing.
- Data access should be limited to the correct authenticated user.
- The platform should be maintainable and scalable for future features.

7. PROTOTYPE STAGE

A prototype is a working representation of the solution. For MemorEase, the prototype is the deployed web application available at the public website link. It demonstrates the brand, landing page, feature previews, help pages, changelog, privacy pages, and authentication flow.

The prototype shows the core identity of the project: study smarter, not harder. The landing page explains that MemorEase transforms study material into cards, reviewers, and exams instantly. This makes the value proposition clear from the first screen.

The visible UI includes a materials hub preview, intelligent content summarization preview, help center, about page, changelog, privacy policy, terms of service, and Google sign-in flow. Together, these screens prove that the project is more than a static idea.

Prototype Architecture

The prototype follows a web application architecture. The frontend displays study workflows and user interface screens. Authentication is handled through Google OAuth and Supabase callback. Study data and generated materials are expected to be stored under authenticated user accounts.

The AI generation layer processes study material and creates learning assets. The security layer validates inputs, controls database access, and protects sign-in from abuse. The deployment layer hosts the application through Vercel.

Component	Prototype Role
Landing Page	Explains the project and encourages students to begin.
Materials Hub	Shows organization of flashcards and reviewers.
AI Summarization	Shows transformation of notes into structured learning content.
Help Center	Documents user-facing features and study tools.
Changelog	Shows development history, features, and security improvements.
OAuth Login	Provides account entry through Google and Supabase.

Security and Privacy Design

The prototype includes visible security-related design choices. The login flow uses Google OAuth through Supabase. The changelog mentions hCaptcha, row-level security, input validation with Zod, rate limiting, CSP, HSTS, and security policies.

The privacy policy explains that account information, uploaded study materials, usage data, preferences, feedback, and communications may be collected for service operation. It also states that third-party services may be used for authentication and AI processing.

For a study platform, these details are important because users upload learning materials and expect their data to remain private. A future production version should continue improving encryption, deletion controls, export options, and transparent AI processing policies.

Figure 7.1 MemorEase Landing Page

The landing page introduces MemorEase with the headline Study smarter not harder. It presents the product as an open-source alternative to Quizlet and Gizmo.

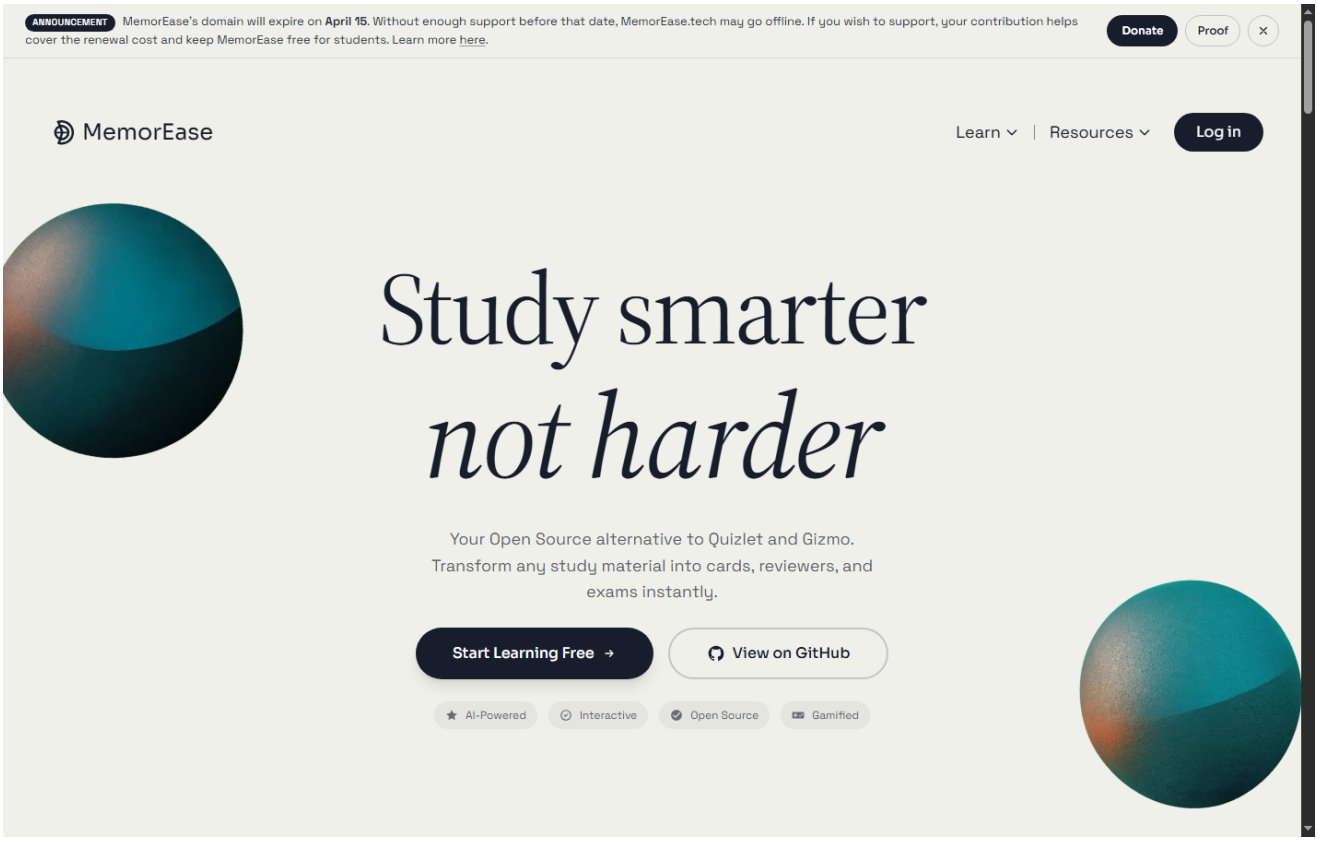


Fig 7.1 MemorEase Landing Page

This screen is important because it immediately communicates the product promise. The Start Learning Free button, GitHub link, and badges such as AI-Powered, Interactive, Open Source, and Gamified show the main identity of the platform.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.2 Materials Hub Preview

The materials hub preview shows saved learning assets such as reviewers and card decks.

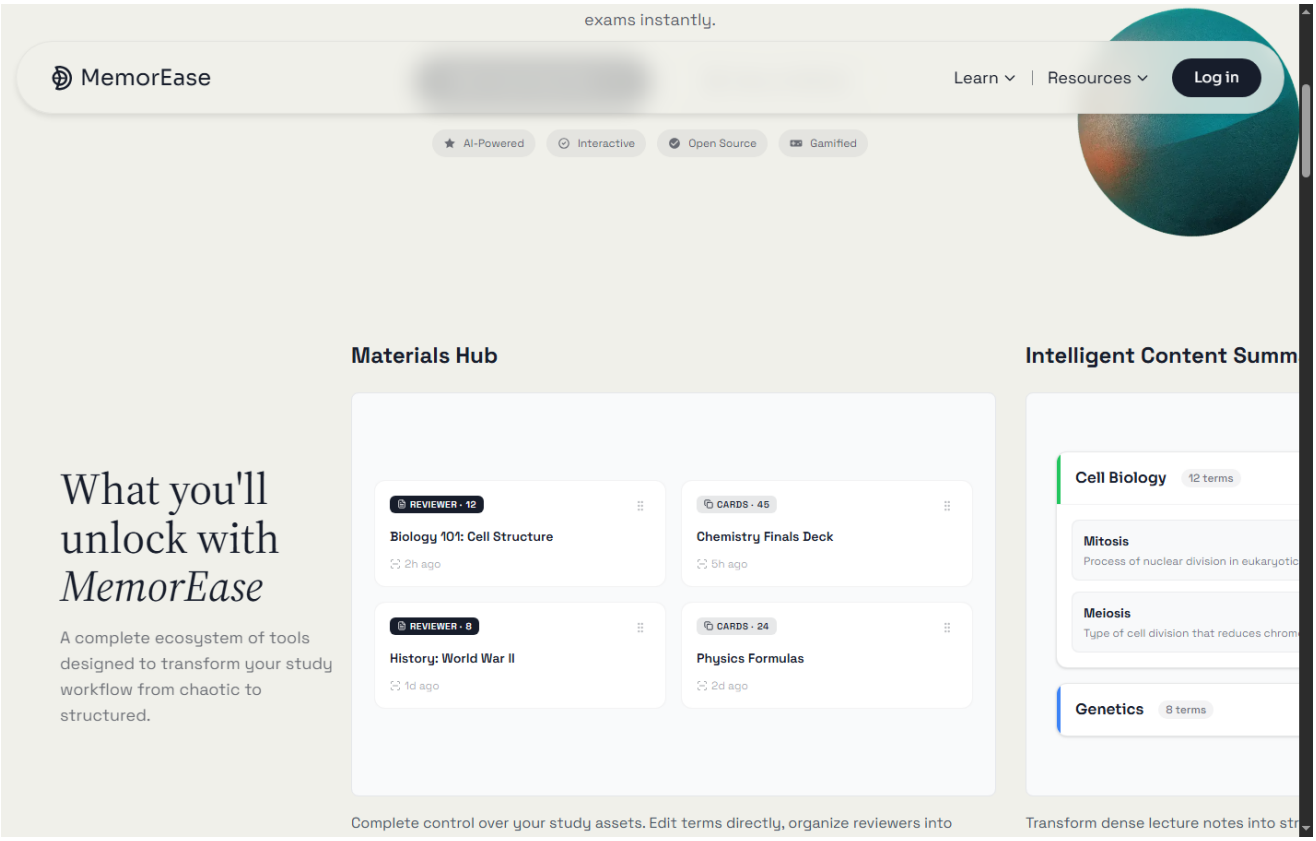


Fig 7.2 Materials Hub Preview

This screen demonstrates how generated study material can be organized. It shows that MemorEase is not only a generator, but also a workspace where students can manage cards and reviewers.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.3 Intelligent Content Summarization

The summarization section shows how a topic such as Cell Biology can be converted into structured terms and definitions.

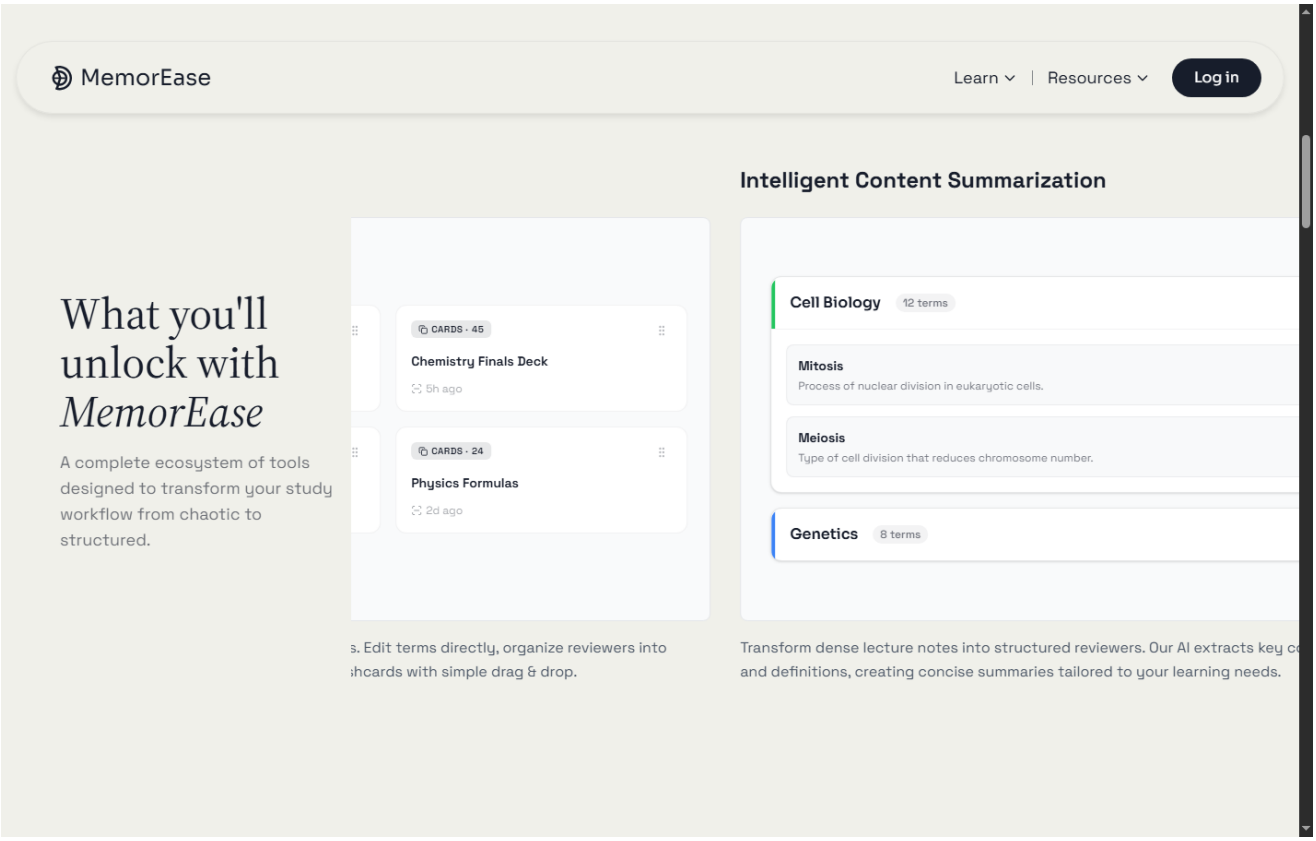


Fig 7.3 Intelligent Content Summarization

This screen connects the AI feature to a real student use case. Dense notes can become readable summaries with concepts such as mitosis, meiosis, and genetics.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.4 Call to Action and Footer

The final landing section invites students to start learning and shows resource links, legal links, and contact options.

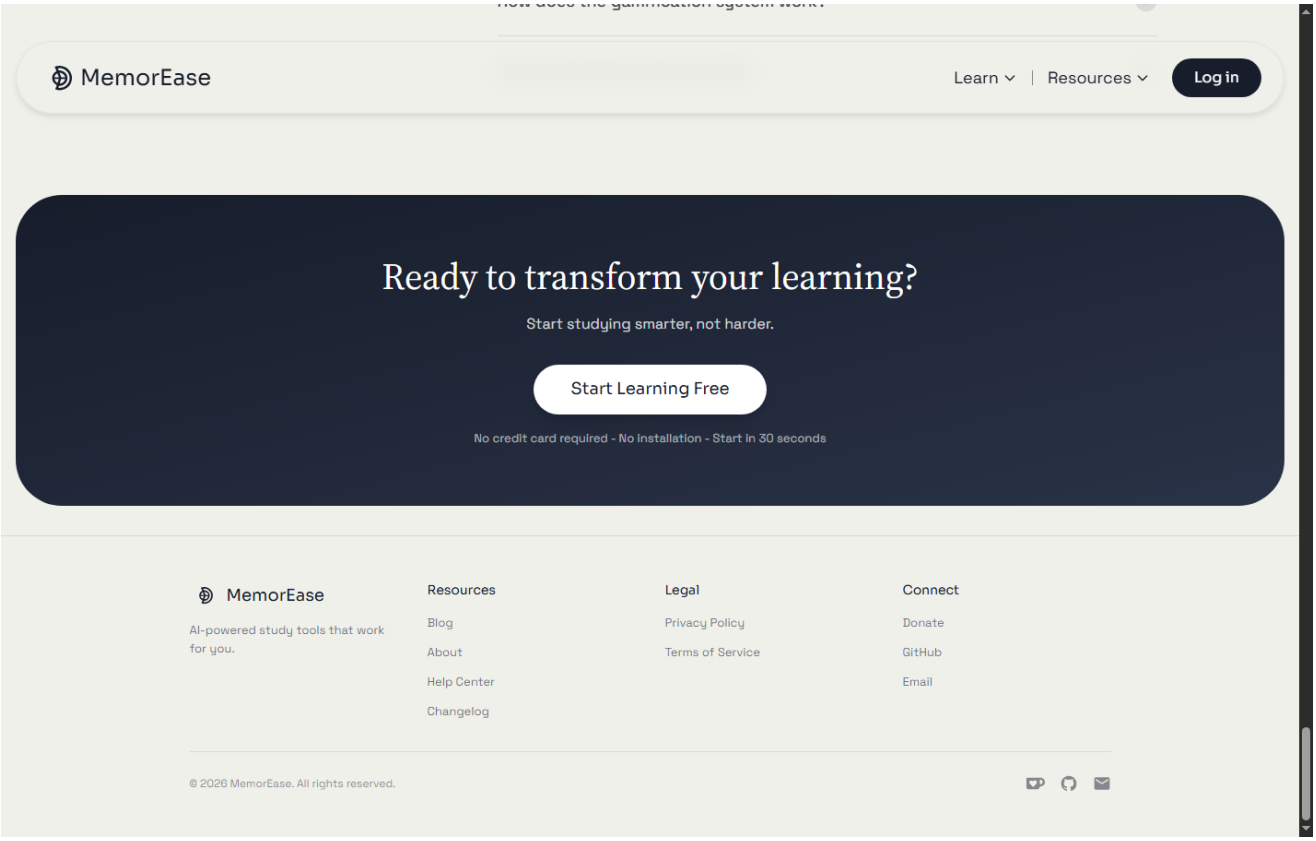


Fig 7.4 Call to Action and Footer

This screen shows project completeness. A real platform needs not only feature screens, but also navigation, support, privacy, terms, GitHub, and contact links.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.5 About Page

The About page explains the mission of MemorEase and why the project was built.

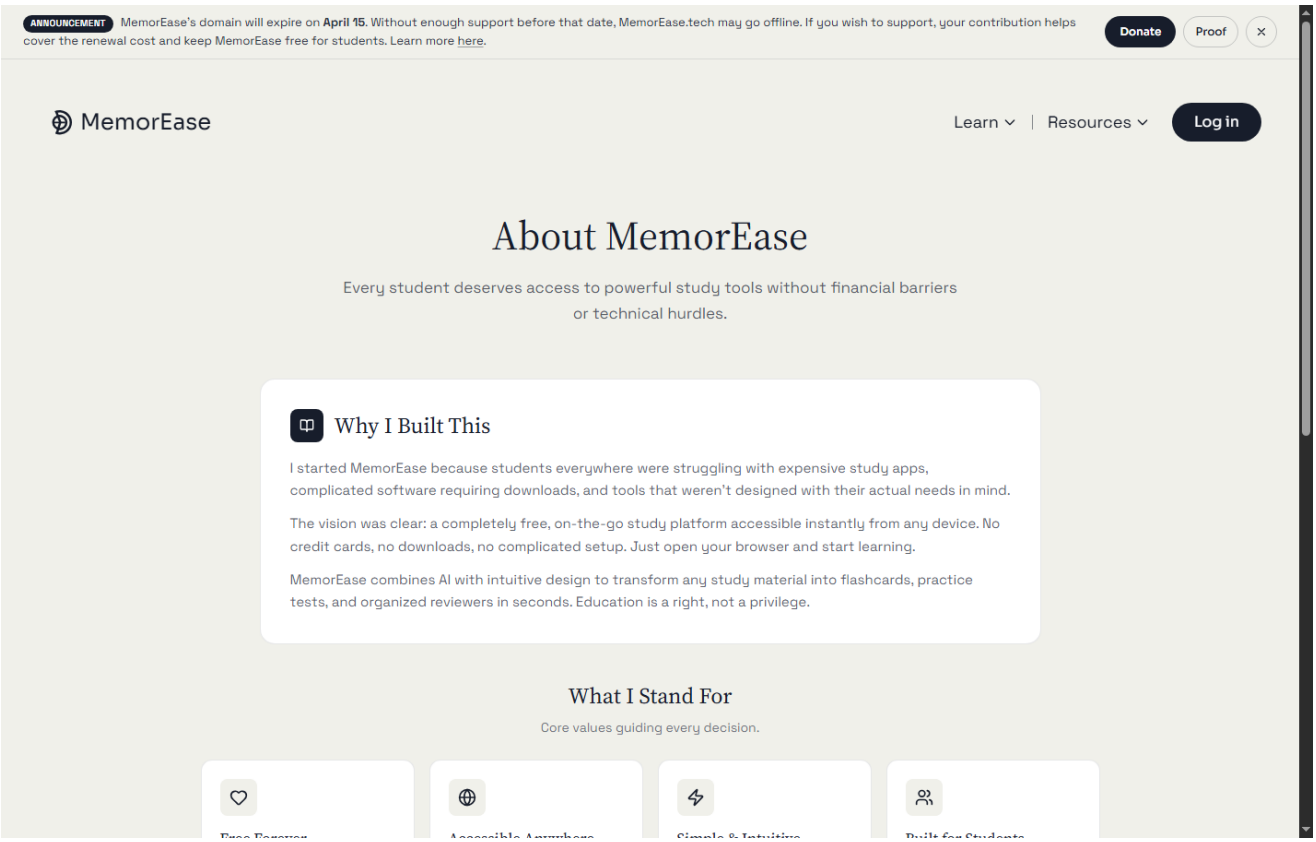


Fig 7.5 About Page

This screen supports the Design Thinking story because it clearly states the student problem: expensive apps, complicated downloads, and tools not designed for actual student needs.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.6 Help Center

The Help Center organizes support topics such as Getting Started, Reviewer, Quiz Mode, Flashcards, Pomodoro Timer, Account Settings, Achievements, Study Calendar, and Leveling System.

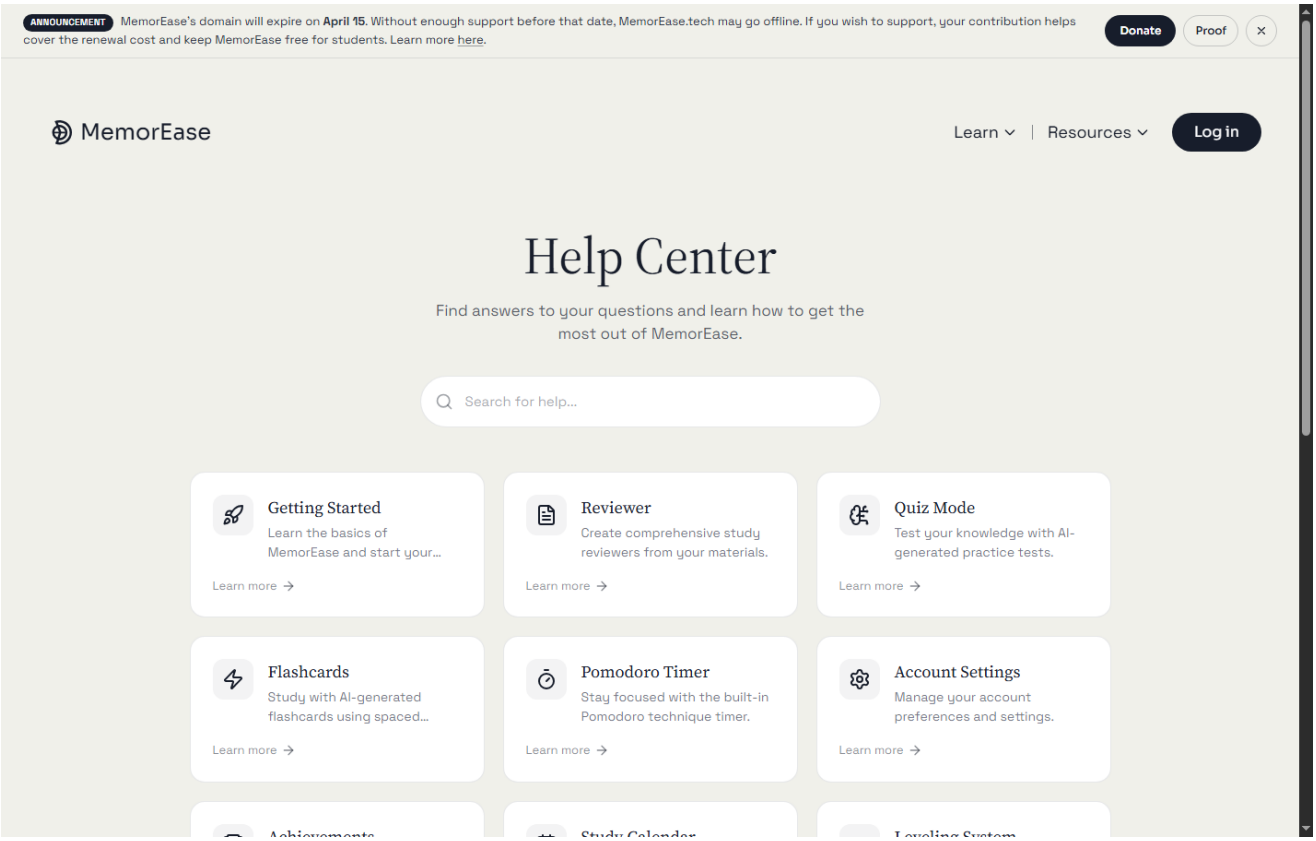


Fig 7.6 Help Center

This screen proves that the project has planned multiple user workflows. It also helps evaluators understand the scope of the platform.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.7 Changelog Page

The changelog lists product versions, feature additions, improvements, and security updates.

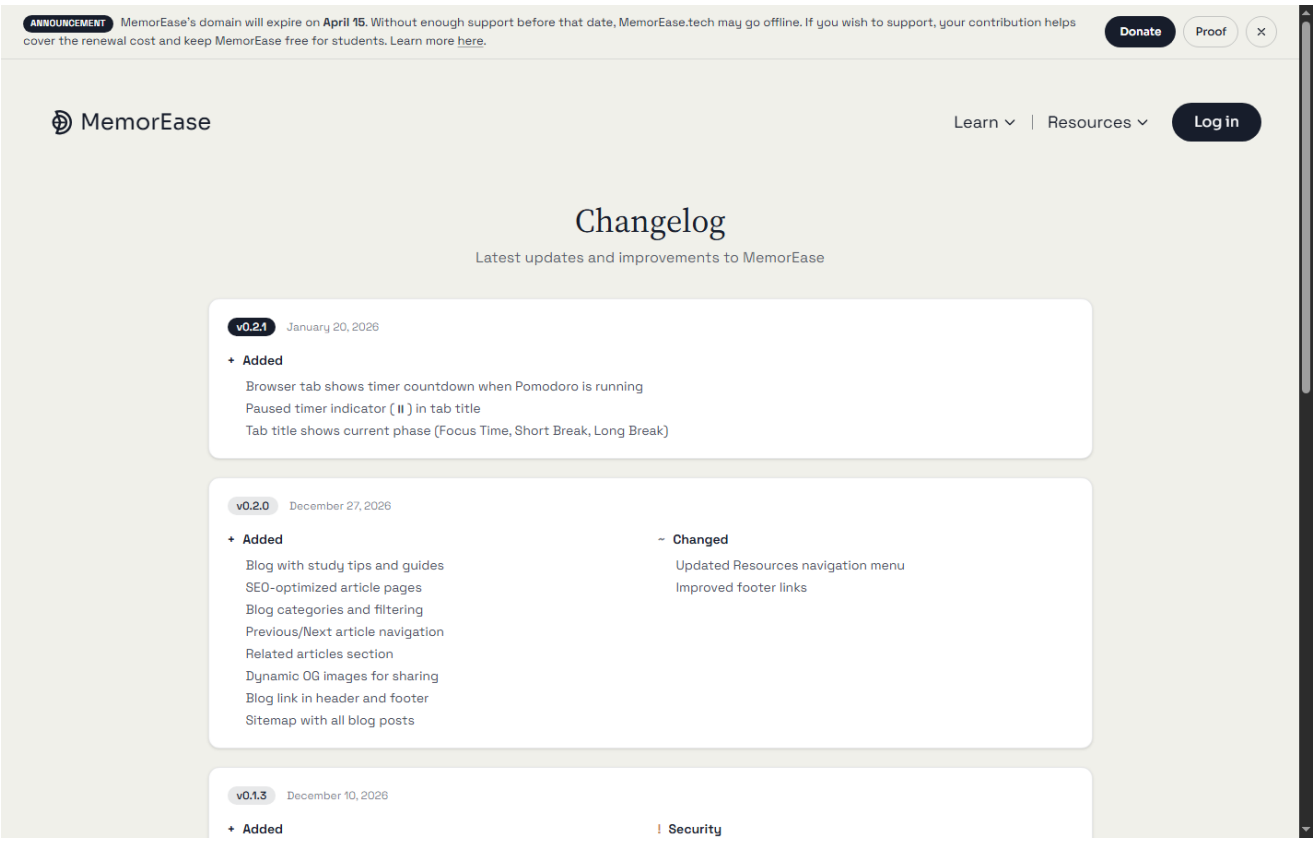


Fig 7.7 Changelog Page

This screen is valuable because it shows that MemorEase is actively developed. It mentions features such as Pomodoro, blog, captcha, RLS, API handling, dashboard accessibility, AI generation, export, and rate limiting.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.8 Privacy Policy

The Privacy Policy explains what information is collected and how data is used.

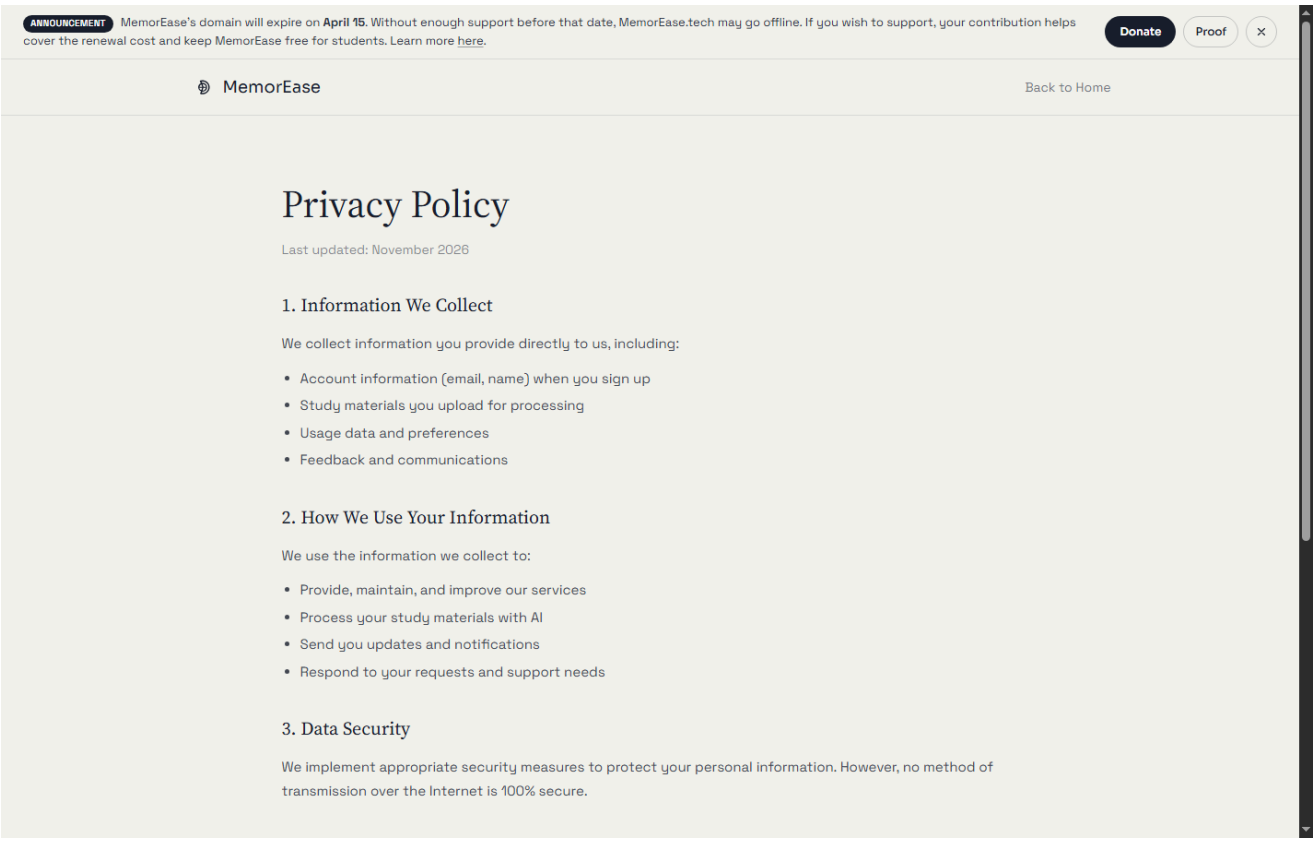


Fig 7.8 Privacy Policy

This screen is important because study tools may process uploaded notes and account information. Privacy documentation improves user trust.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.9 Terms of Service

The Terms page describes the service, acceptable use, ownership of uploaded content, analytics, and user responsibilities.

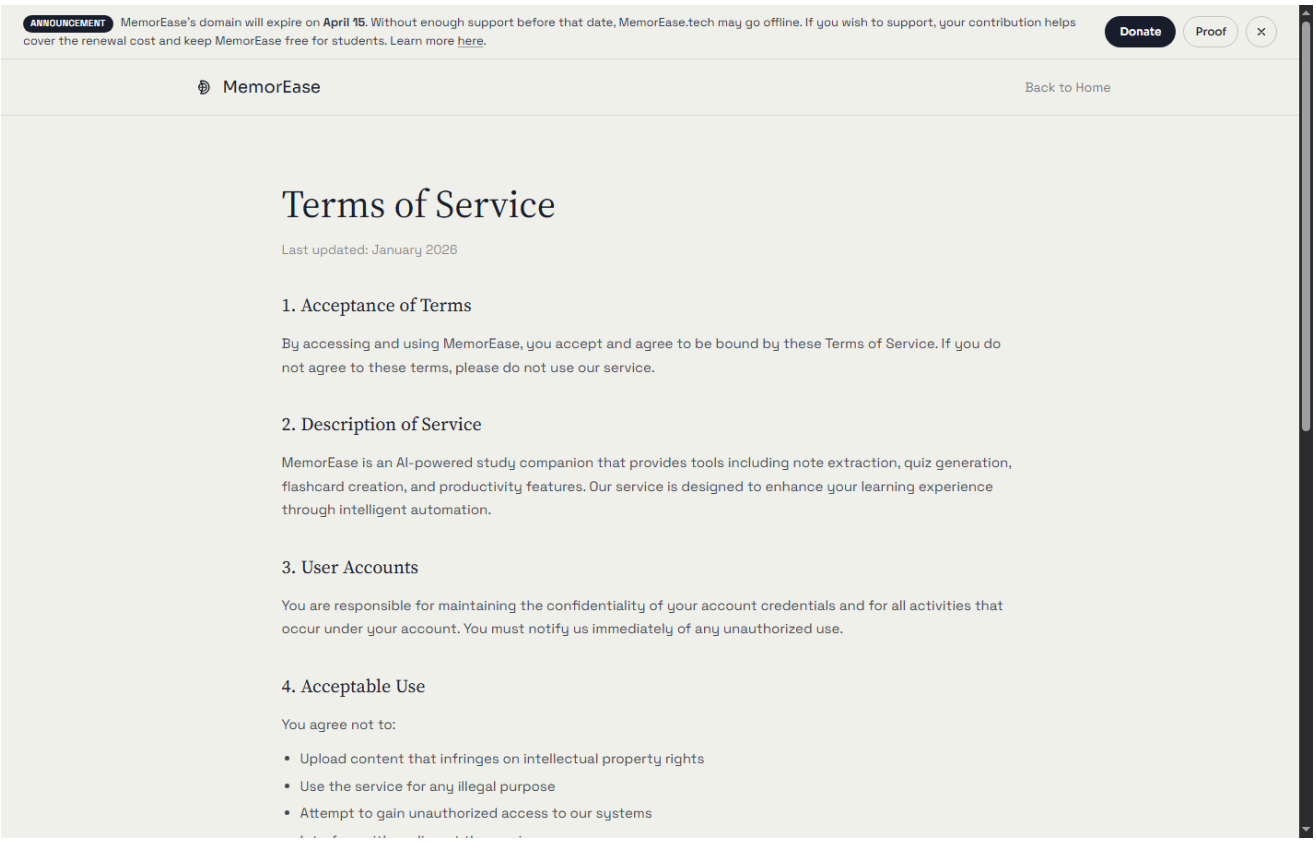


Fig 7.9 Terms of Service

This screen makes the prototype look closer to a real product because it includes legal and usage expectations.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.10 Google OAuth Login

The login screen appears when the user selects Log in or Start Learning Free.

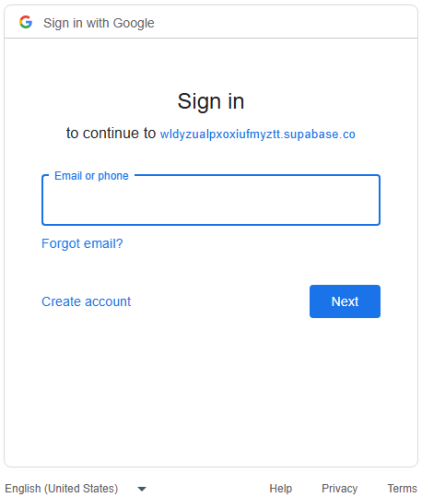


Fig 7.10 Google OAuth Login

This screen shows that authentication is not just planned. The prototype connects to Google sign-in through a Supabase callback, supporting account-based study storage.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.11 Mobile Responsive Home Screen

The mobile screenshot shows how the landing page adapts to a smaller screen.

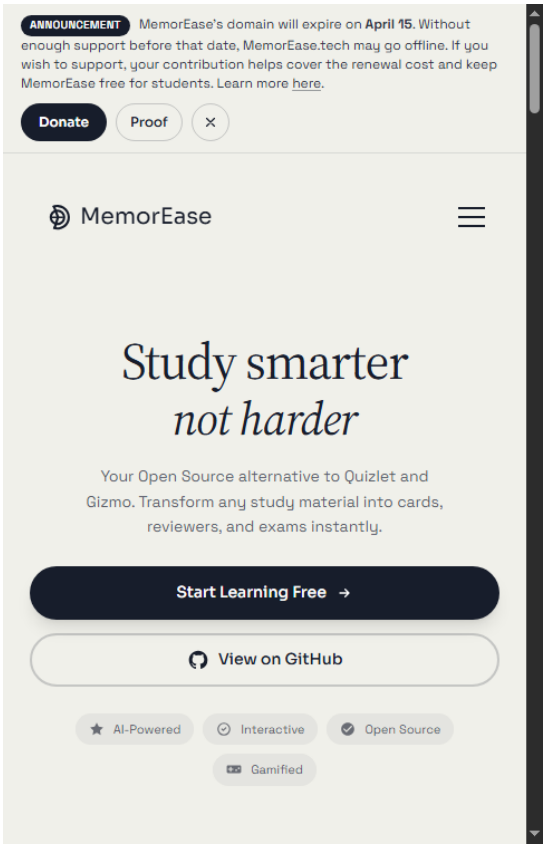


Fig 7.11 Mobile Responsive Home Screen

This screen demonstrates responsive design. Since students may study on phones, mobile readability is important for the success of MemorEase.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.12 Logged-in Home Dashboard

The logged-in home dashboard shows the learner greeting, level status, XP progress, today's study time, current streak, best streak, recent activity, and study calendar.

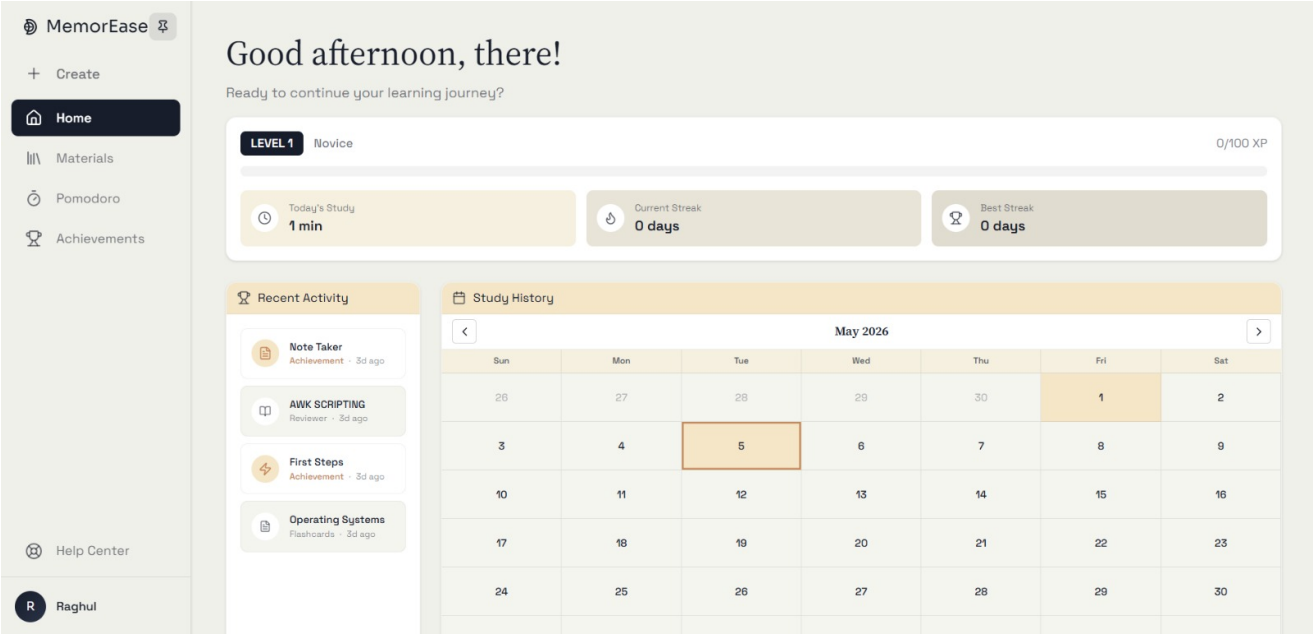


Fig 7.12 Logged-in Home Dashboard

This screen is important because it shows MemorEase as a daily learning workspace. The student can quickly understand progress, recent materials, achievements, and calendar activity without opening many separate pages.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.13 Create Materials Page

The Create Materials page allows the student to choose between Cards and Reviewer, enter a material title, and select Manual Entry, Bulk Add, or AI Generate.

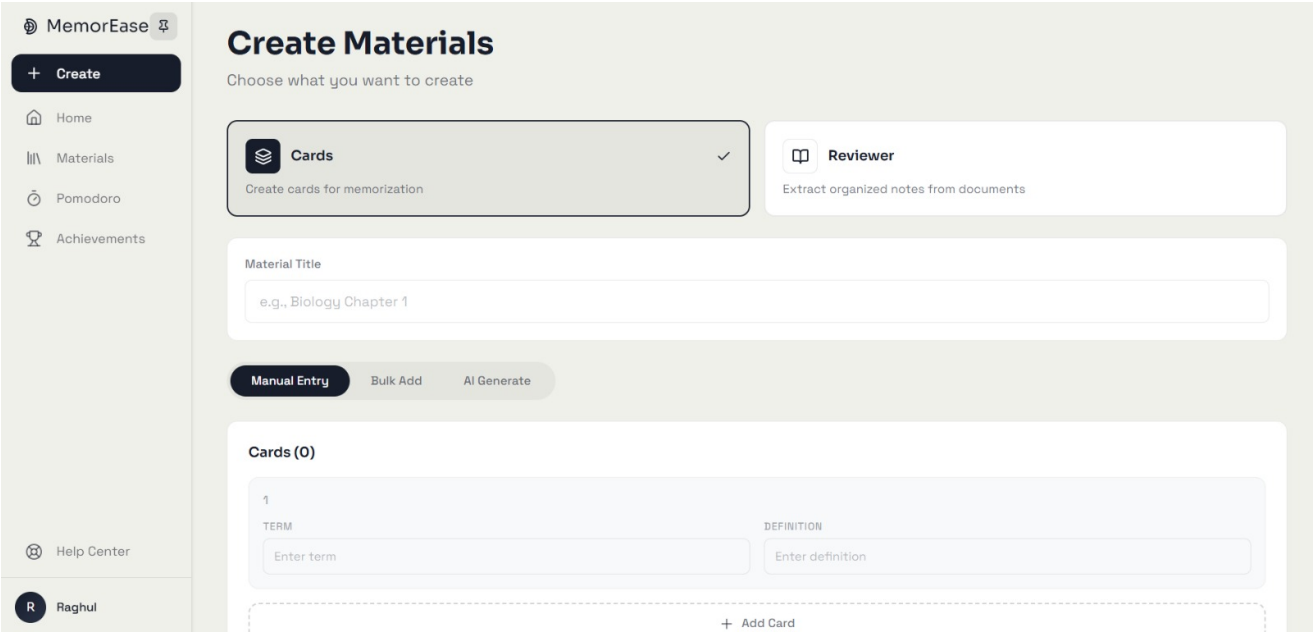


Fig 7.13 Create Materials Page

This screen explains the content creation workflow. It shows that MemorEase supports both simple manual card creation and future AI-assisted generation, which makes the platform useful for different study habits and different types of notes.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.14 Materials Library

The Materials page shows saved study assets with a search bar and filters for All Items, Cards, and Reviewer.

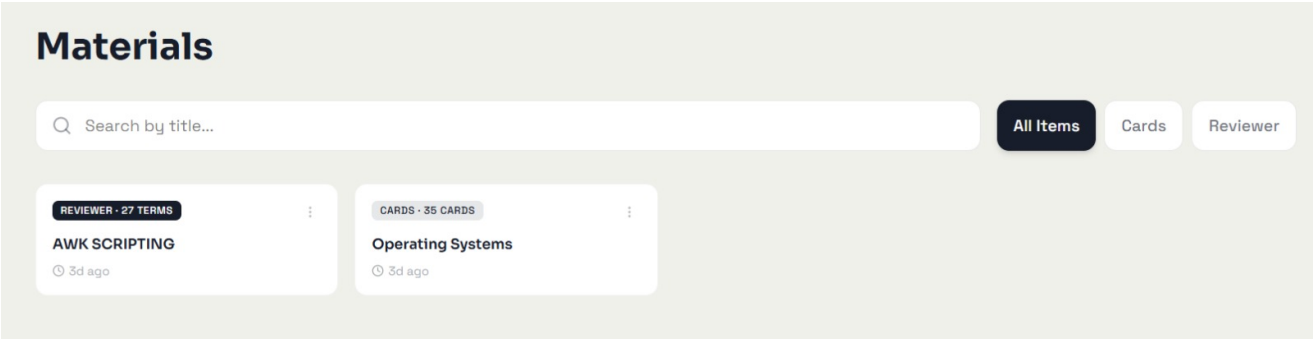


Fig 7.14 Materials Library

This screen demonstrates organization. Students can keep multiple subjects or chapters in one place, search by title, and quickly identify whether an item is a reviewer or a flashcard deck.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.15 Material Detail and Study Modes

The Operating Systems material page shows Flashcards, Learn, Practice, and Match modes along with study progress and a list of terms.

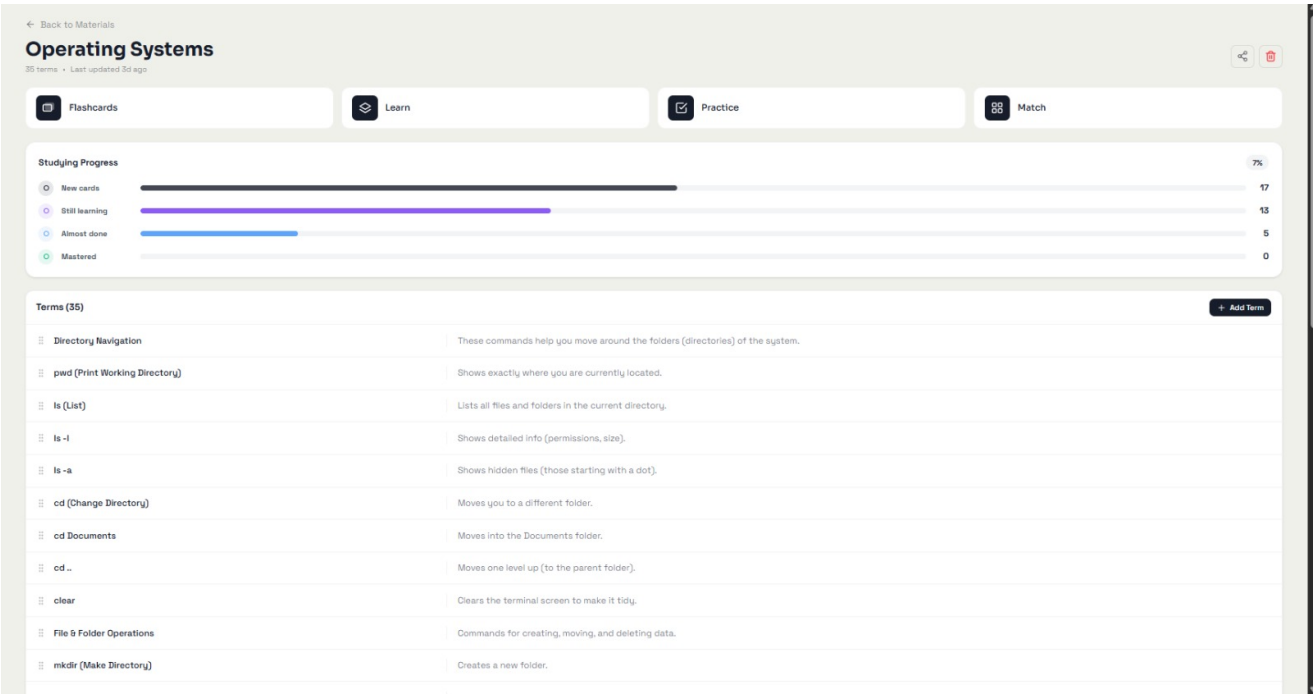


Fig 7.15 Material Detail and Study Modes

This screen is one of the strongest prototype screens because it connects stored material with active learning. The progress bars show new, still learning, almost done, and mastered states, helping students understand what remains to be revised.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website

separately.

Figure 7.16 Pomodoro Timer and Task Panel

The Pomodoro screen provides focus time, short break, long break, timer controls, and a task panel.

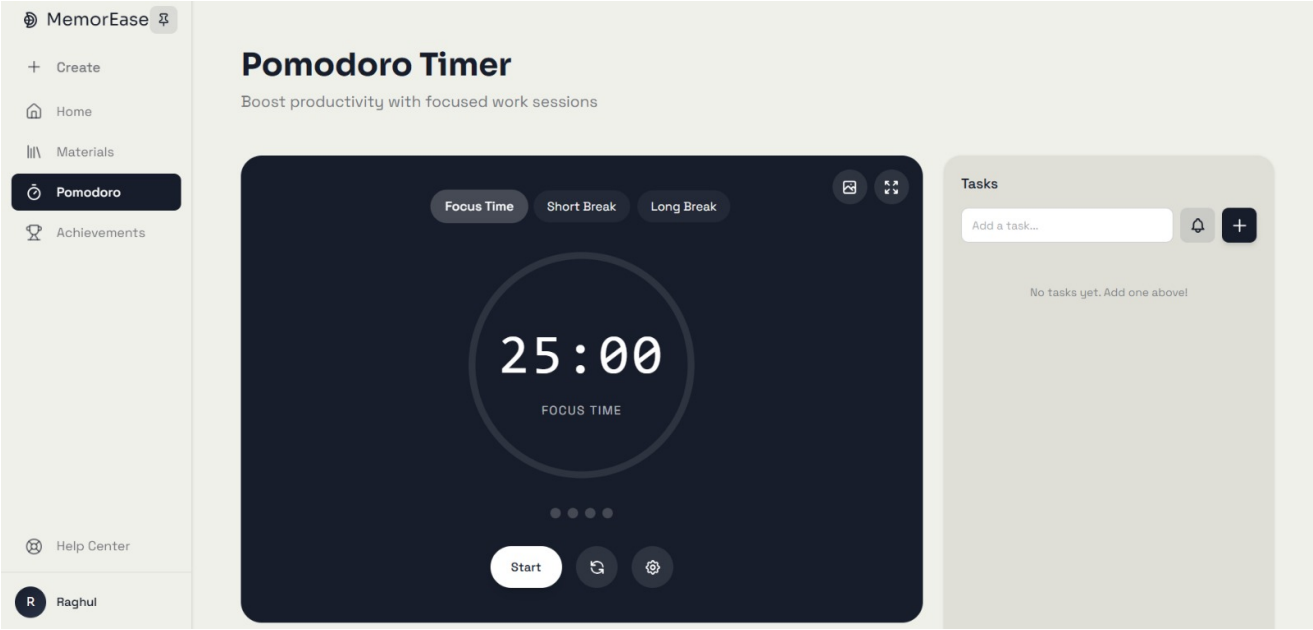


Fig 7.16 Pomodoro Timer and Task Panel

This screen adds productivity support to the study platform. Students can revise using timed sessions, add tasks, and maintain focus instead of switching to a separate timer application.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.17 Achievements Dashboard

The Achievements page shows unlocked and locked achievements, overall progress, and rewards such as First Steps, Note Taker, Memory Starter, and Flashcard Master.

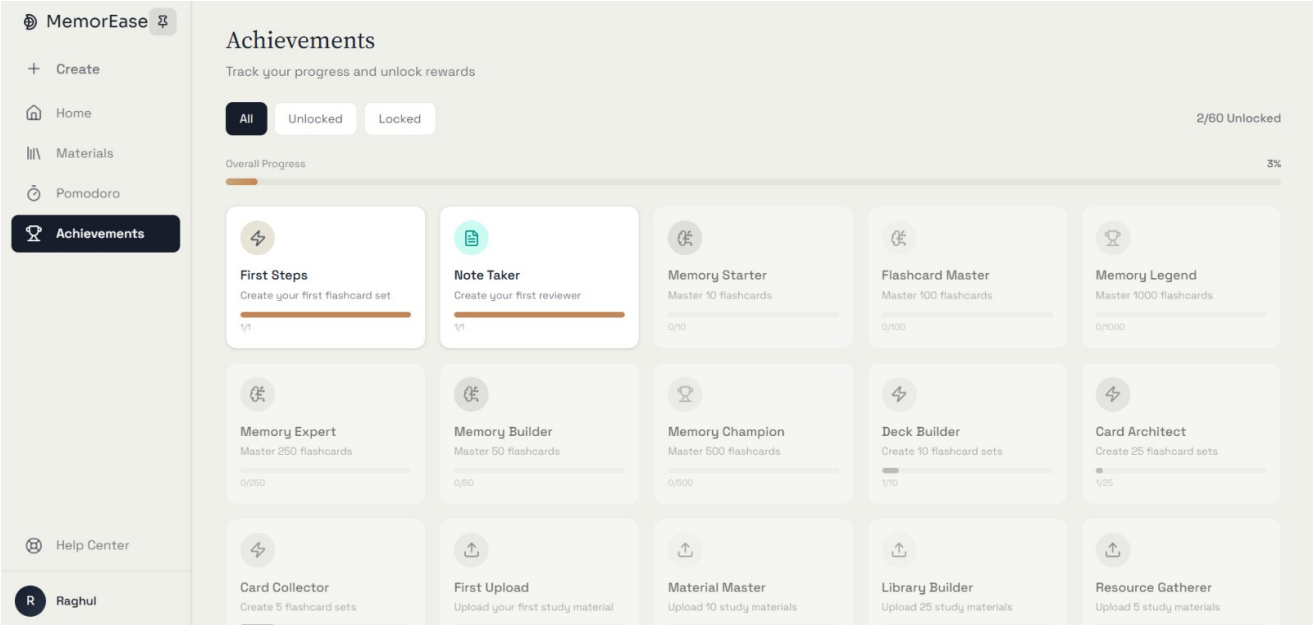


Fig 7.17 Achievements Dashboard

This screen shows the gamification layer of MemorEase. Achievements encourage repeated usage by turning study progress into visible goals, which can improve motivation during exam preparation.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.18 Practice Options Modal

The Practice Options modal lets the user choose the number of questions, enable question types, shuffle terms, show answer feedback, and auto-advance after answers.

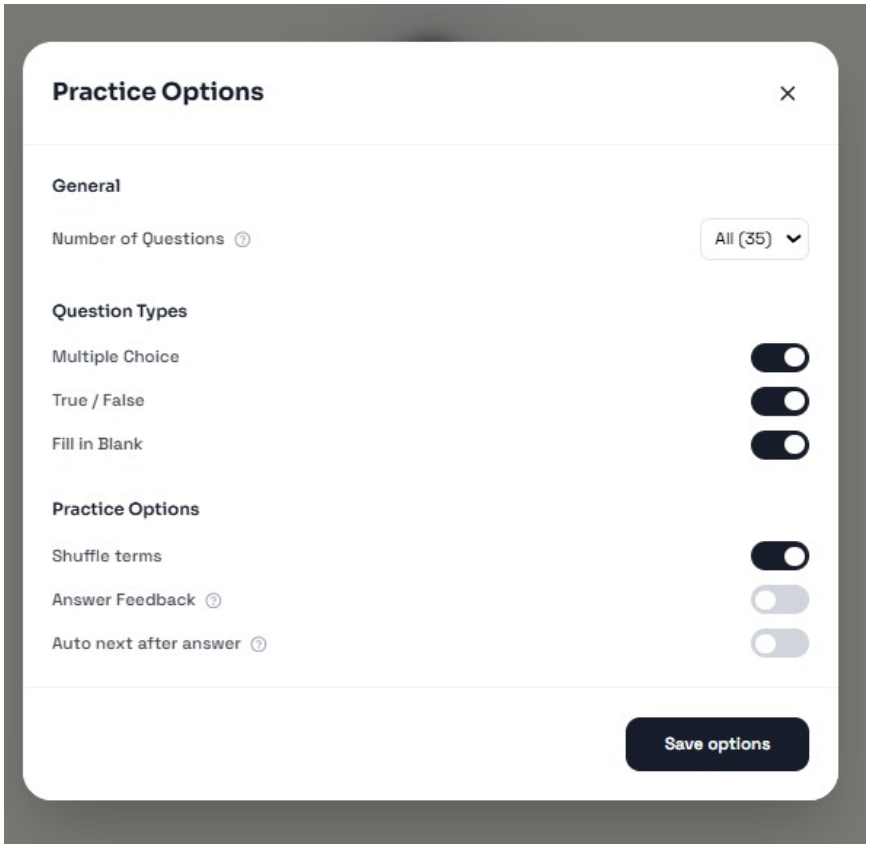


Fig 7.18 Practice Options Modal

This screen is useful because it gives the learner control over the test experience. Students can adjust practice sessions based on time, confidence, and exam preparation needs.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.19 Practice Question Screen

The practice screen presents a definition prompt, progress indicator, answer input, skip option, answer button, and next action.

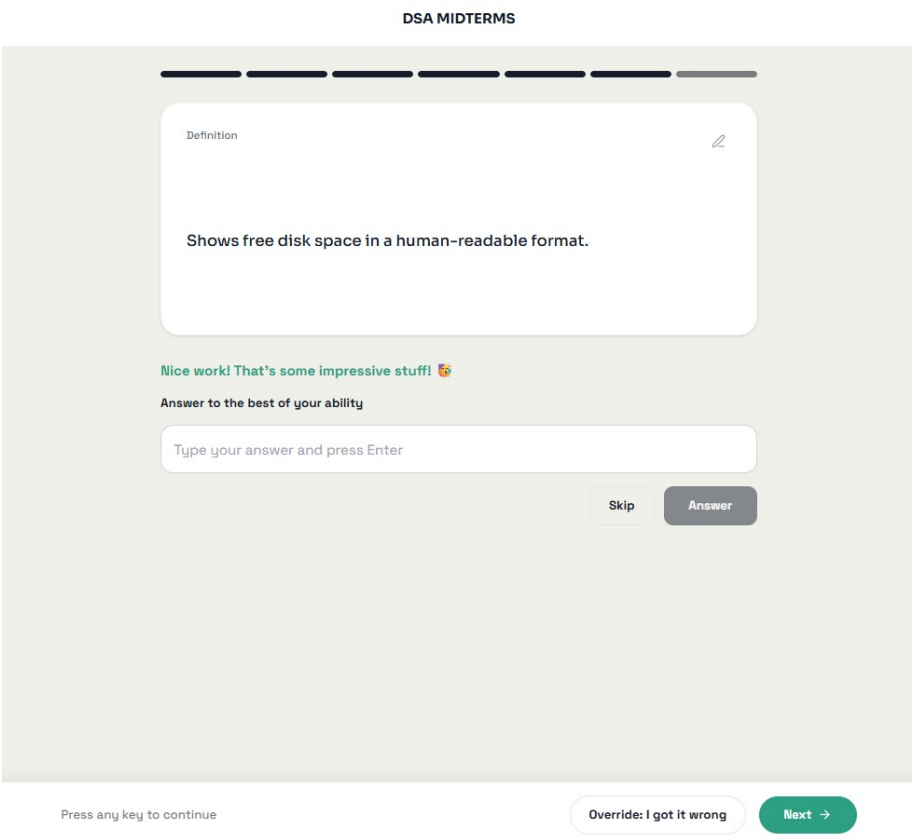


Fig 7.19 Practice Question Screen

This screen demonstrates active recall. Instead of simply reading notes, the student must remember and type an answer, making the learning process more effective and measurable.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.20 Progress Warning Modal

The progress warning modal appears when the user tries to leave a study session and explains that unsaved progress may be lost.

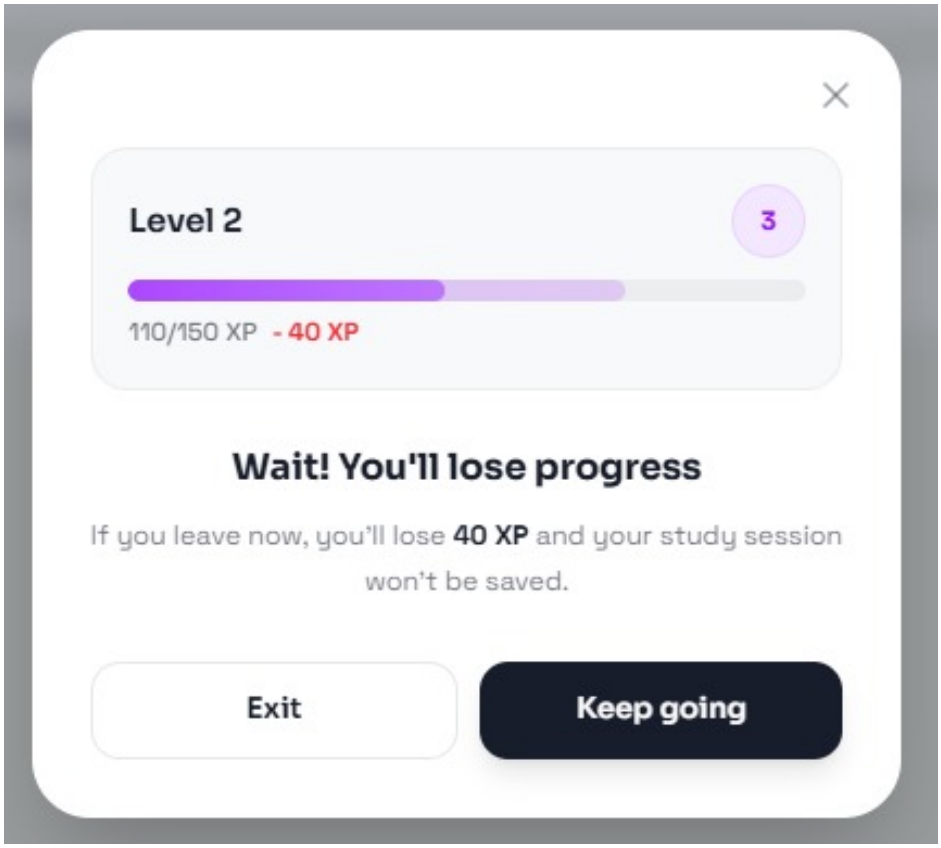


Fig 7.20 Progress Warning Modal

This screen improves usability by preventing accidental loss of work. It also shows that the prototype considers real user behavior during study sessions, where students may accidentally close or leave a page.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Figure 7.21 Share Material Modal

The Share Material modal allows the user to create a share link with an optional custom code.

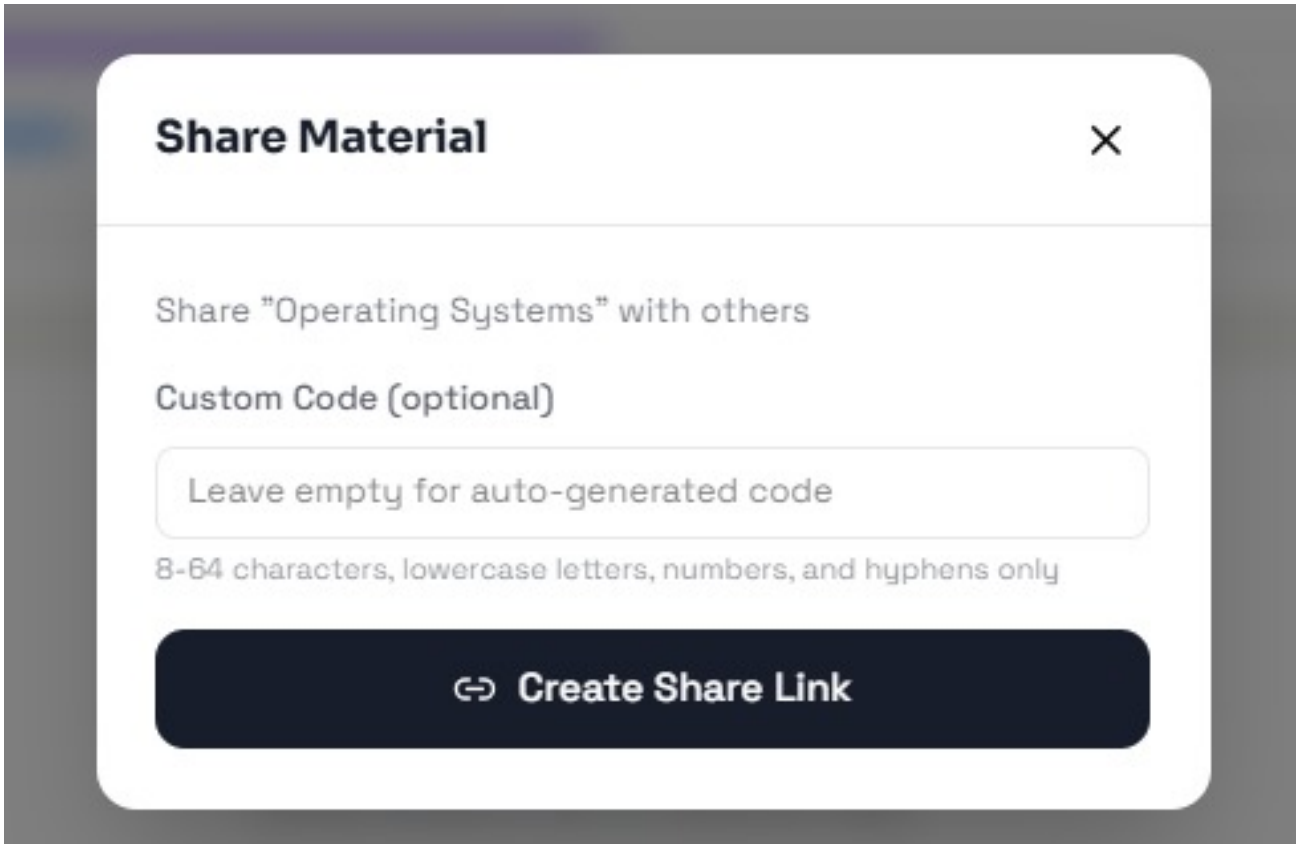


Fig 7.21 Share Material Modal

This screen supports collaboration. A student can share a deck or reviewer with classmates while keeping the sharing action controlled through a generated link or custom code.

In the report, this screen is included to connect the written concept with the actual prototype. It shows how the idea is converted into a usable interface, which is an important part of the Prototype stage in the Design Thinking process. The screenshot also helps evaluators understand the navigation flow without opening the website separately.

Prototype Screen Summary

The UI screenshots show that MemorEase has a complete product direction. The landing page attracts users, the preview sections explain learning features, the About page explains mission, the Help Center documents workflows, the Changelog proves development, and the login screen demonstrates authentication.

The UI also supports trust. Privacy Policy, Terms of Service, support links, GitHub links, and changelog entries make the platform feel more realistic and transparent.

The prototype section is important in the Design Thinking process because it converts the idea into something visible and testable. Students and evaluators can understand the project without reading only technical explanation.

A strong point of the prototype is that it communicates both learning value and product credibility. The homepage explains the benefit, the Help Center explains available tools, and the legal pages explain how the service should be used. This combination makes the project appear closer to a deployable educational product.

The prototype also shows that MemorEase is designed for repeated use. A student is expected to return to the platform for flashcards, quizzes, Pomodoro sessions, achievements, and calendar tracking. This repeated-use design is important because learning improvement happens over time.

Another important observation is that the public pages use a calm and readable visual style. The interface avoids unnecessary complexity and gives attention to headings, cards, spacing, and simple actions. This is suitable for students because the tool should reduce study stress, not add more confusion.

Overall, the prototype validates the main concept of the project. It shows that AI study generation, structured materials, authentication, support content, privacy awareness, and responsive design can be combined into one student-focused platform.

The landing screen maps to the awareness stage of the user journey. A student first understands what MemorEase does and why it is useful. The feature preview screens then move the student from awareness to interest by showing materials, reviewers, cards, and summaries.

The About and Help pages support the evaluation stage of the journey. Before using a new learning tool, students may want to know why it was built, what features exist, and whether it matches their study habits. These pages answer those questions directly.

The Privacy, Terms, and OAuth login screens support the trust stage. A study platform handles uploaded material and account information, so users need to see clear policies and secure sign-in before they begin saving personal study content.

Therefore, the prototype screens form a complete public journey: understand the product, inspect the features, check support information, review trust documents, and

sign in to begin learning. This makes the UI section important evidence for the project report.

The prototype also helps identify what should be tested next. Since the public screens already explain the idea well, the next testing focus should be on the logged-in dashboard, AI generation quality, card editing, quiz accuracy, and whether students can complete a full study session without confusion.

From a report perspective, these UI screens prove that the project has crossed the planning stage. The website is deployed, navigable, responsive, and connected to authentication. This gives the project stronger evidence than a purely theoretical proposal.

The next chapter therefore evaluates the prototype through functional and usability testing. The testing section checks whether the visible system behavior supports the stated goals of accessibility, clarity, student usefulness, and trust.

The prototype also gives a clear basis for future internal testing. Once a user signs in, the most important flows to test are material creation, AI generation, card review, quiz submission, export, Pomodoro completion, and progress update. These flows represent the practical core of the platform.

Testing should also check whether a first-time student can complete the journey without guidance. The ideal path is simple: understand the landing page, sign in, add material, generate study assets, revise, test knowledge, and track progress.

This transition from prototype to testing is central to Design Thinking. A prototype is not successful only because it looks good; it becomes successful when users can use it, understand it, and give feedback that improves the next version.

8. TESTING AND FEEDBACK

The prototype was tested through scenario-based review. The test focused on whether a new user could understand the platform purpose, navigate the website, identify main features, open support pages, and reach the login flow.

The main evaluation areas were clarity, navigation, feature communication, accessibility, security visibility, and student usefulness. Since the authenticated internal dashboard could not be accessed without a real account, the test focused on public prototype screens and visible workflows.

Test Case	Action	Expected Result	Status
TC-01	Open landing page	User understands MemorEase as an AI study platform.	Pass
TC-02	View feature preview	Materials hub and AI summarization are visible.	Pass
TC-03	Open About page	Mission and student problem are explained.	Pass
TC-04	Open Help Center	Study features are listed as support topics.	Pass
TC-05	Open Changelog	Version updates and security improvements are visible.	Pass
TC-06	Open Privacy / Terms	User can read data and service policies.	Pass
TC-07	Click Login	Google OAuth sign-in screen opens.	Pass
TC-08	Check mobile view	Landing page remains usable on small screen.	Pass

Feedback Interpretation

The landing page communicates the main idea clearly. The phrase Study smarter not harder is easy to understand and matches the purpose of reducing study preparation effort.

The feature previews are helpful because they show examples instead of only text. The materials hub and summarization screens allow users to imagine how the platform will organize their own study content.

The Help Center improves confidence because it lists major tools such as flashcards, quiz mode, Pomodoro, achievements, study calendar, and leveling system. This shows that the project has a wider learning ecosystem.

The Changelog is especially useful for evaluation because it gives evidence of real development. It shows security improvements, feature additions, API handling, export support, and study mode expansion.

One improvement area is to make the authenticated dashboard preview more visible to users who are not logged in. A public demo mode or sample account would help evaluators understand the internal study workflow better.

Usability Evaluation

- Clarity: The website communicates the project purpose quickly.
- Navigation: Public pages such as About, Help, Changelog, Privacy, and Terms are easy to access.
- Usefulness: Flashcards, reviewers, quizzes, Pomodoro, and gamification match student needs.
- Trust: OAuth, privacy policy, terms, changelog, and security notes improve confidence.
- Responsiveness: The mobile screen shows that the interface adapts to smaller devices.
- Improvement: A public demo dashboard can make the prototype easier to evaluate.

9. RE-DESIGN AND IMPLEMENTATION

Based on prototype testing, the design can be improved by making the student journey more visible before login. The landing page already communicates the idea well, but a demo dashboard or sample material flow would help new users understand the full product without creating an account.

The Help Center can be expanded with step-by-step screenshots for creating flashcards, generating reviewers, starting quiz mode, and using Pomodoro. This would make onboarding easier for first-time users.

The Changelog indicates that security and accessibility improvements have already been considered. Future redesign should continue improving dashboard accessibility, mobile layouts, and clear feedback after AI generation.

Area	Improvement
Landing Page	Keep the strong value statement and add a public demo preview.
Materials Hub	Show sample edit, reorder, and category actions more clearly.
AI Generation	Add status messages, editable output, and confidence warnings.
Study Modes	Expose flashcards, learn, match, and practice mode previews.
Security	Continue RLS, validation, captcha, and privacy documentation.
Mobile UX	Improve compact navigation and study controls for phone users.

10. RESULT AND CONCLUSION

MemorEase successfully demonstrates the concept of an AI-powered study companion. The deployed website communicates a clear product vision: students can transform study material into flashcards, reviewers, and exams instantly.

The project solves an important student problem by reducing manual preparation time and organizing revision into active learning formats. It supports educational methods such as active recall, spaced repetition, practice testing, and focused study sessions.

The prototype also shows real product maturity through support pages, changelog, privacy policy, terms, OAuth login, mobile responsiveness, and feature descriptions. These elements make the project stronger than a simple landing page.

The additional logged-in screenshots further prove that the platform has a complete learning workflow. The student can create materials, manage cards and reviewers, open a subject, choose different study modes, practise questions, use a Pomodoro timer, track achievements, and share materials with classmates.

The result of the project is therefore not only a static website, but a structured educational web application. It combines content organization, AI-assisted material creation, active recall, gamification, focus management, and user progress tracking in one interface.

From a Design Thinking point of view, MemorEase satisfies the main problem statement. Students need a simple and affordable way to convert notes into revision-ready learning assets. The prototype answers this need through a clean dashboard, clear material flow, useful study modes, and supportive productivity features.

Although the platform can still be improved with stronger AI verification, mobile application support, teacher mode, and advanced analytics, the current version already validates the main concept. With further development, MemorEase can become a practical learning companion for students across different academic levels.

11. WRITE UPS

Project Write Up

MemorEase is an AI-powered study companion created to support students in organizing, revising, and practising academic content. The project focuses on a common problem faced by learners: study material is available in large quantities, but it is not always available in a form that is easy to revise. Students often have long notes, copied definitions, textbook paragraphs, PDF files, and last-minute exam topics. Reading all of this repeatedly takes time and does not always improve memory.

The aim of MemorEase is to convert this scattered study material into useful learning formats such as flashcards, reviewers, practice questions, and structured study resources. A student can create materials manually, add content in bulk, or use AI generation to reduce the effort required for preparing revision cards. The system is designed as a browser-based platform so that students can use it without installing a heavy application.

The project is meaningful because it connects artificial intelligence with real student needs. Many AI tools only generate text, but MemorEase tries to place AI inside a complete study workflow. The generated content is not the final goal; it becomes part of a larger learning process where the student reviews cards, practises questions, tracks progress, and builds consistency.

MemorEase is also designed around accessibility. The platform presents itself as a free and open-source alternative to paid study applications. This is important because students may not always be able to pay for premium learning tools. By offering a free study companion, the project supports equal access to digital learning resources.

User Interface Write Up

The user interface of MemorEase is clean, calm, and study-focused. The dashboard greets the learner and shows progress details such as level, XP, streak, study time, recent activity, and calendar history. This gives the student a quick overview of learning consistency. Instead of hiding progress inside many menus, the home screen makes important study information visible immediately.

The Create Materials page is one of the central screens of the project. It allows the user to choose between Cards and Reviewer. It also provides a material title field and

tabs such as Manual Entry, Bulk Add, and AI Generate. This design is useful because not all students prepare in the same way. Some prefer entering definitions manually, some want to paste many terms at once, and some may want AI help.

The Materials page works as the library of the platform. Students can search by title and filter content by All Items, Cards, or Reviewer. Each material appears as a card with type and update information. This helps the user manage multiple subjects and chapters. Without this library structure, generated content would become difficult to find later.

The material detail screen shows the deeper study experience. The Operating Systems example includes Flashcards, Learn, Practice, and Match modes. It also shows progress categories such as new cards, still learning, almost done, and mastered. This turns learning into a trackable process rather than a one-time reading activity.

The Pomodoro screen supports focused study sessions. The timer provides focus time, short break, and long break modes, while the task panel lets students add small work goals. This is useful because students often lose focus when switching between study tools. MemorEase keeps the timer inside the same learning environment.

The Achievements screen adds motivation. It shows unlocked and locked rewards such as First Steps, Note Taker, Memory Starter, and Flashcard Master. These small goals encourage students to continue using the platform. Gamification is useful when it supports learning behavior without distracting from the main study task.

Feature Write Up

The practice features are important because they support active recall. In the practice screen, the student reads a definition prompt, enters an answer, and moves to the next question. This is more effective than passive reading because the learner must remember the concept from memory. The progress bar also gives a sense of completion during the session.

The Practice Options modal improves flexibility. Students can choose the number of questions, enable multiple choice, true or false, and fill in blank modes, shuffle terms, and control answer feedback. This makes the practice session adaptable to different preparation stages. Before an exam, a student may want all questions, while during a quick revision session, a smaller set may be enough.

The progress warning modal is a small but important usability feature. It warns the user before leaving a session where progress may be lost. This prevents accidental mistakes and shows that the system considers real student behavior. During studying, interruptions are common, so clear warnings protect user effort.

The Share Material modal supports collaboration. Students often prepare with classmates, and a useful deck or reviewer can help more than one person. By allowing a share link with an optional custom code, MemorEase can support peer learning while keeping sharing controlled.

Design Thinking Write Up

In the Empathize stage, the project focuses on student difficulties such as exam pressure, lack of time, expensive study apps, unorganized notes, and weak revision habits. These problems are practical and easy to observe in college learning environments. Students do not simply need more content; they need better ways to remember and practise existing content.

In the Define stage, the problem is framed clearly: students need a free, simple, and intelligent platform that can convert study material into revision-ready learning assets. This problem statement helps avoid unnecessary features and keeps the project centered on learning improvement.

In the Ideation stage, several solution ideas are considered, including AI-generated flashcards, reviewer extraction, quiz mode, Pomodoro timer, achievements, calendar tracking, shareable materials, and progress analytics. These ideas work together because they cover preparation, practice, focus, motivation, and review.

In the Prototype stage, the deployed MemorEase website and the logged-in UI screens represent the solution. The prototype shows how the idea becomes a real interface. Students can see the dashboard, materials, creation flow, study modes, practice controls, Pomodoro timer, and achievements. This validates the product direction visually and functionally.

In the Testing stage, the prototype can be evaluated through student scenarios. A user should be able to sign in, create material, add cards, open a deck, practise questions, use the timer, and view progress. Feedback from these scenarios can guide the next version of the platform.

Technical Write Up

Technically, MemorEase can be implemented as a modern web application with a frontend for user interaction, a backend or API layer for processing requests, a database for storing users and study materials, and an AI service for generating cards or reviewers. Authentication can be handled through Google OAuth or a similar secure login provider.

The database should store user accounts, materials, cards, reviewer content, practice attempts, achievements, Pomodoro sessions, and share links. Each record must be connected to the correct user so that private study material is not exposed to other accounts. Row-level security or strong backend authorization is important for this type of platform.

The AI generation flow should be handled carefully. Uploaded or typed content should be validated, processed, and converted into draft cards or notes. Students should be allowed to edit generated output before saving because AI may sometimes produce incomplete or inaccurate content. Human control is important in an education platform.

The future technical version can include adaptive spaced repetition, offline export, mobile application support, teacher dashboards, classroom sharing, and improved analytics. These improvements can make MemorEase more useful in real academic environments while keeping the current student-first design.

Write Up Area	Explanation
Project Purpose	To convert study material into flashcards, reviewers, practice questions, and organized learning resources.
Main Users	Students who need quick revision, better memory support, and affordable study tools.
Core Workflow	Create material, generate or enter cards, review content, practise questions, track progress, and repeat revision.
UI Strength	Clean dashboard, materials library, study modes, Pomodoro timer, achievements, and sharing screens.
Future Direction	Adaptive learning, AI quality control, mobile app, teacher mode, collaboration, and deeper analytics.

12. FUTURE WORK

- Add a public demo dashboard with sample materials and study modes.
- Improve AI quality checking and allow users to edit generated cards before saving.
- Add advanced spaced repetition scheduling based on card difficulty and review history.
- Develop mobile app versions for Android and iOS.
- Add collaborative study groups where classmates can share decks with permission.
- Add offline mode or exportable study packs for low-connectivity students.
- Improve analytics with retention score, weak topics, daily streaks, and exam readiness.
- Add voice notes and audio-based revision for auditory learners.
- Add teacher mode for creating class materials and monitoring student progress.
- Strengthen privacy controls with clearer data deletion, export, and AI processing options.

Future Scope Explanation

The future scope of MemorEase is strong because the current prototype already contains the base learning workflow. The next step is to make the system more intelligent and more personalized. Instead of only storing cards and reviewers, the platform can observe student performance and suggest which topics should be revised next.

A major future improvement is adaptive spaced repetition. The system can track whether the student answered a card correctly, how long the student took, and how many times the same concept was missed. Based on this data, difficult cards can appear more often while mastered cards can appear less often.

Another important improvement is AI quality control. Generated cards should be treated as editable drafts. The platform can include warnings, confidence indicators, source previews, and user editing tools. This will make AI support more reliable for academic use.

Collaboration can also make the platform more useful. Students can share materials with classmates, create group decks, and revise common subjects together. In future

versions, teachers can create verified decks for an entire class and monitor general progress without exposing private student data.

Mobile access is also important. Many students revise on phones while travelling, waiting between classes, or preparing quickly before tests. A mobile app or improved progressive web app can make card review, Pomodoro, and practice sessions easier on small screens.

These future features can make MemorEase more powerful and more useful in real academic environments. The current prototype proves the concept, while future versions can deepen AI personalization, collaboration, analytics, and mobile access.

13. LEARNING OUTCOMES OF DESIGN THINKING

- Design Thinking helped convert a broad education problem into a focused student-centered platform.
- The Empathize stage showed that students need speed, simplicity, affordability, and motivation.
- The Define stage helped frame the problem as unorganized study material and lack of accessible tools.
- The Ideation stage encouraged AI flashcards, reviewers, quiz mode, Pomodoro, achievements, and calendar features.
- The Prototype stage showed how a deployed website can communicate the product vision.
- The Testing stage showed that public pages, feature previews, and login flow are understandable.
- The project improved understanding of AI in education, authentication, privacy, UI design, and scalable web applications.

Learning Outcome Explanation

The project helped in understanding how a broad idea can be shaped into a clear product. At the beginning, the problem was simply that students need help while studying. Through Design Thinking, this broad need became a specific solution: a study companion that converts material into flashcards, reviewers, practice sessions, and progress tracking.

The Empathize stage showed the importance of observing real user pain points. Students do not always struggle because they lack content. Many times they struggle because content is unorganized, revision takes too long, and paid tools are not affordable. Understanding this changed the project direction from a simple AI generator to a complete study workflow.

The Define stage improved problem framing. Instead of trying to build every possible education feature, the project focused on the key need of turning study material into revision-friendly formats. This made the system easier to explain and easier to evaluate.

The Ideation stage encouraged multiple connected features. Cards, reviewers, practice mode, Pomodoro, achievements, sharing, and calendars were not selected randomly. Each feature supports a different part of the learning process: preparation, revision, testing, focus, motivation, and collaboration.

The Prototype stage helped convert written ideas into visible screens. UI screenshots made the report stronger because they show how users will actually interact with the product. The dashboard, material creation page, study modes, practice options, and achievements show that the solution has a practical interface.

The Testing stage showed that a project should be judged by usability, clarity, and usefulness, not only by technical features. A student should be able to understand the platform quickly and complete a study session without confusion. Feedback from testing can guide redesign and future improvements.

Overall, the project improved practical knowledge of web application planning, user interface design, database thinking, authentication, AI integration, privacy, and educational technology. It also showed how sustainability in software can mean making learning resources more accessible and affordable.

Additional Technical Notes

14. REFERENCES

[1] MemorEase deployed website. Available: https://memor-ease-chi.vercel.app/

[2] MemorEase About page. Available: https://memor-ease-chi.vercel.app/about

[3] MemorEase Help Center. Available: https://memor-ease-chi.vercel.app/help

[4] MemorEase Changelog. Available: https://memor-ease-chi.vercel.app/changelog

[5] MemorEase Privacy Policy. Available: https://memor-ease-chi.vercel.app/privacy-policy

[6] MemorEase Terms of Service. Available: https://memor-ease-chi.vercel.app/terms

[7] H. L. Roediger and J. D. Karpicke, Test-enhanced learning: Taking memory tests improves long-term retention, Psychological Science, 2006.

[8] P. C. Wozniak, Optimization of learning, SuperMemo spaced repetition research resources.

[9] Supabase Documentation. Available: https://supabase.com/docs

[10] Vercel Documentation. Available: https://vercel.com/docs

[11] Google Identity OAuth Documentation. Available: https://developers.google.com/identity/protocols/oauth2

[12] Google Gemini API Documentation. Available: https://ai.google.dev/docs

[13] Zod Documentation. Available: https://zod.dev/

[14] PostHog Documentation. Available: https://posthog.com/docs